Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

# SIDE KICK

## Geography

Topographical maps, map of India and every chapter of the 2027 syllabus, with practice questions and a marking checklist.



EVERY TOPPER HAS A SIDEKICK

saviours.pro

# Hey. I'm your Sidekick.

I live inside this book. Wherever things get tricky, I pop up with a tip, a trap to dodge, or a question the board loves to ask. Read the chapter, then try my HOT questions. Deal?



**Aligned to the official ICSE 2027 syllabus. If it's in the syllabus, it's in here.**

**SIDEKICK TIP**
Shortcuts, mnemonics and exam hacks.

**FUN FACT**
Things that make the concept stick.

**REAL LIFE**
Where this shows up outside the exam hall.

**KEY POINT**
Definitions and laws you must write exactly.

**HOT QUESTION**
Higher-order thinking. Try before you peek.

**✦ NEW**
Added in this edition to match the 2027 syllabus.

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01

CHAPTER · GEOGRAPHY

# Interpretation of Topographical Maps

A topographical sheet is a bird's-eye view of a small piece of India drawn to scale. Learn its language of grids, brown lines, colours and symbols and this section becomes the most predictable part of the paper.



IN THIS CHAPTER

- Grid references
- Contours and landforms
- Colours and symbols
- Scale and area
- Directions
- Drainage and settlements

## What Is a Topographical Map?

A **topographical map** (toposheet) shows natural and man-made features of a small area in detail: relief, rivers, forests, villages, roads, railways. In India the maps are produced by the **Survey of India**. The sheet in your exam is usually drawn at **1:50,000** and printed with a rectangular **grid** of numbered lines.



FUN FACT

Sheets follow a family tree. A 1:1,000,000 sheet (say number 45) splits into 16 sheets of 1:250,000 (lettered A to P), and each of those into 16 sheets of 1:50,000 (numbered 1 to 16). That is why a code looks like 45D/10.

## 1. Locating Features by Grid Reference

Vertical lines are **eastings** and horizontal lines are **northings**. Each line has a number printed in the margin. They cut the sheet into squares, and in a 1:50,000 sheet each square is **1 km by 1 km**.

**1. Four-figure reference**

Name the square by the easting then the northing of its **bottom-left (south-west)** corner. Example: 2342.

**2. Six-figure reference**

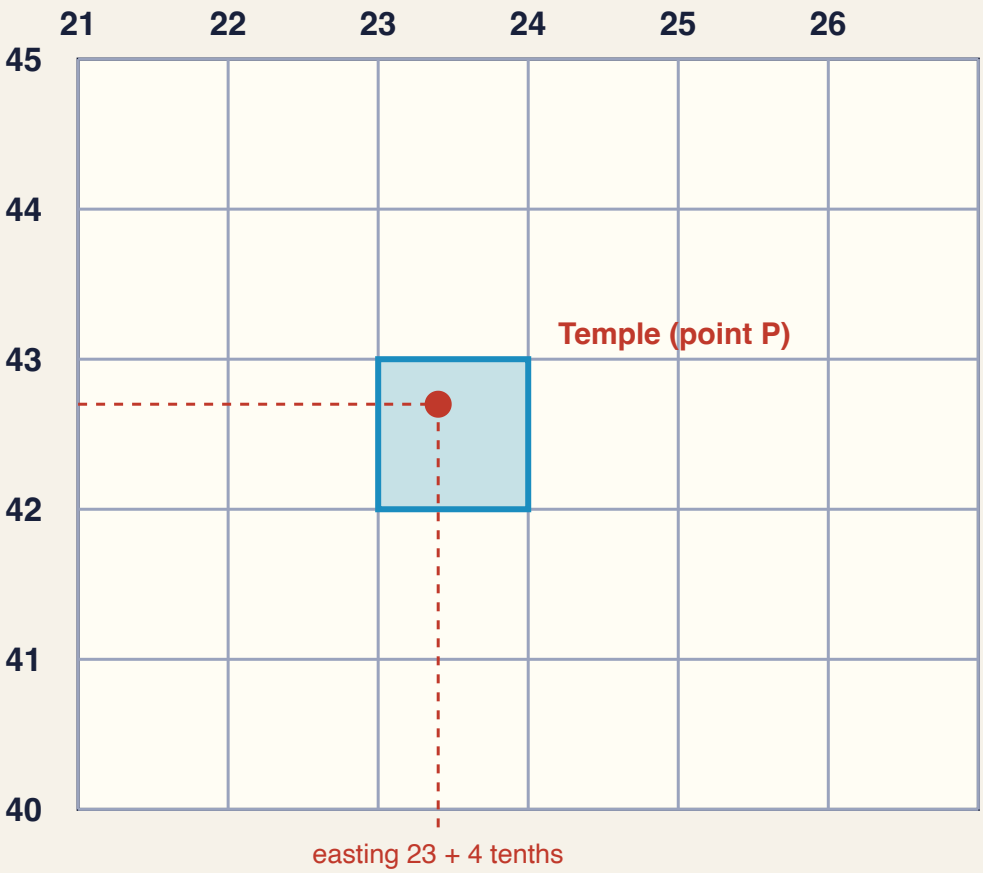
Add a third digit to each number. Divide the square into tenths along each side and read how many tenths east of the easting line and north of the northing line. Example: 234427 means 23 + 4 tenths east, 42 + 7 tenths north.

**3. Golden rule**

Eastings first, northings second: **along the corridor, then up the stairs**.

**4. Checking**

A four-figure answer fixes an area of 1 sq km; a six-figure answer fixes a point to within about 100 m.




**4-figure reference of P: 2342 | 6-figure reference of P: 234427**

Eastings are read first (left to right), then northings (bottom to top).

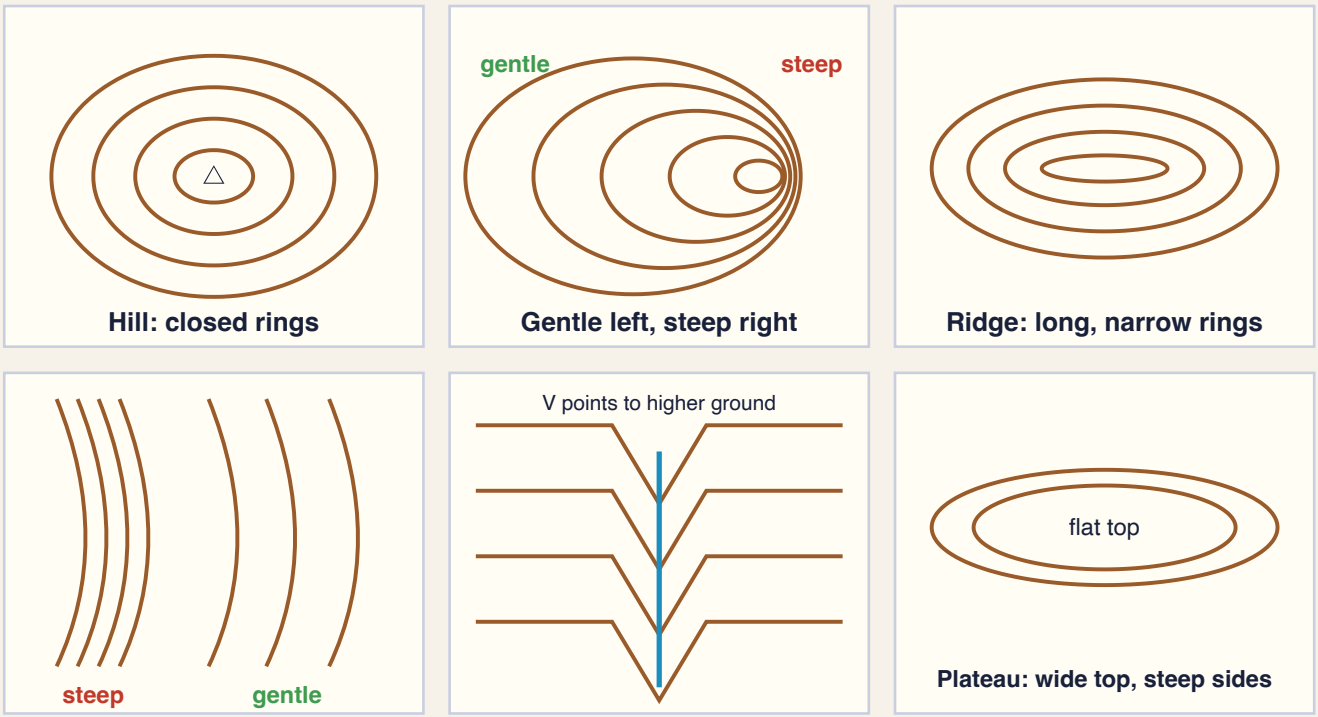
BOARD ALERT

Never swap the order. 4223 and 2342 point to different places. Also read the margin numbers of the sheet itself, never guess them from a diagram.

## 2. Contours and Relief

A **contour** is an imaginary line on the map joining places of **equal height above mean sea level**. The fixed height difference between two neighbouring contours is the **contour interval** (often 20 m on a 1:50,000 sheet). Contours are drawn in **brown** and every fifth line is usually thicker with its value printed on it.

| SPACING AND SHAPE OF CONTOURS | LANDFORM |
|---|---|
| Close together | **Steep slope** |
| Wide apart | **Gentle slope** |
| Concentric closed rings, height rising inwards | **Hill** |
| Long narrow closed rings | **Ridge** |
| Close on one side, wide on the other | **Escarpment**: steep on one face, gentle on the other |
| V-shapes with the point toward higher ground | **Valley** with a stream (a spur is the opposite: the V points down-slope) |
| Wide ring with steep sides and a flat top | **Plateau** |




Brown lines are contours. Spacing tells the story: close = steep, wide = gentle.

### Heights on the Map

| TERM | SYMBOL | MEANING |
|---|---|---|
| Spot height | A black dot with a number, e.g. •352 | Height of a point found by levelling. Not on a contour line |
| Triangulated height | A black triangle with a number, e.g. △467 | Height fixed by a trigonometrical survey, more exact, usually on a hilltop |
| Bench mark (BM) | BM with a number, e.g. BM 245.6 | Permanent mark on a bridge or building with a known height |
| Relative height | Difference of two heights | Height of one point above another. Hilltop 467 m and village 392 m gives 75 m |



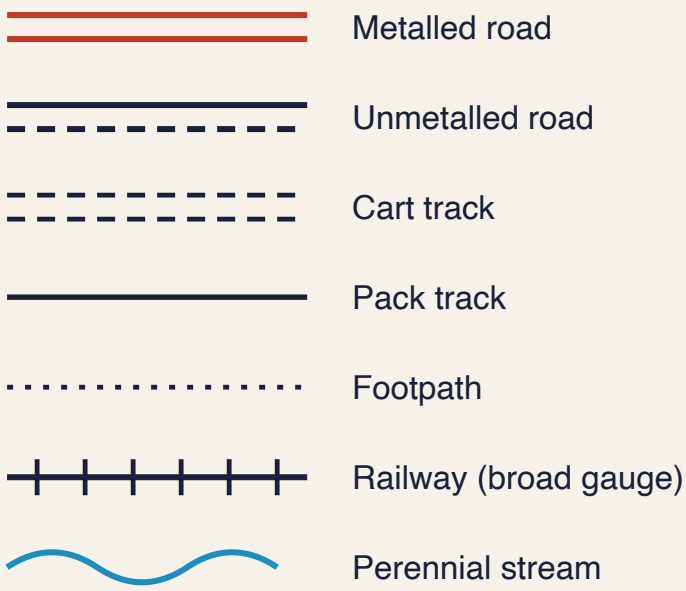
SIDEKICK TIP

Relative **depth** works the same way for a hollow: subtract the lower value from the higher one. Always write the unit (metres).

## 3. Colours and Conventional Symbols

Colour does half the reading for you. Learn the set once.

| COLOUR | WHAT IT SHOWS |
|---|---|
| **Blue** | Water: rivers, streams, tanks, lakes, wells |
| **Brown** | Relief: contours and sand dunes |
| **Green** | Vegetation: forest (dark green), orchards and grass (lighter shades) |
| **Yellow** | Cultivated land |
| **White** | Uncultivated or barren land and snow |
| **Red** | Settlements, main roads and built-up areas |
| **Black** | Names, grid lines, railways and other man-made features |




How roads, tracks and a railway are drawn (typical Survey of India style).

A few short forms appear all over the sheet. They are explained in the index at the bottom, so check it before you answer.

| SHORT FORM | MEANING | SHORT FORM | MEANING |
|---|---|---|---|
| PO | Post office | PS | Police station |
| Sch | School | RH | Rest house |
| DB | Dak bungalow | Hosp | Hospital |
| Temp | Temple | Mos | Mosque |

## Roads, Tracks and Services

Use the index to name the means of transport. A **metalled road** is a hard-surfaced road for all seasons; an **unmetalled road** is only laid with gravel or mud; a **cart track** is wide enough for a bullock cart; a **pack track** is a narrow hill trail for mules and ponies; a **footpath** is for walkers only. Railways are black lines with cross-bars, and their gauge is shown on the key. **Communication** includes post and telegraph offices, and services include schools, hospitals and rest houses.

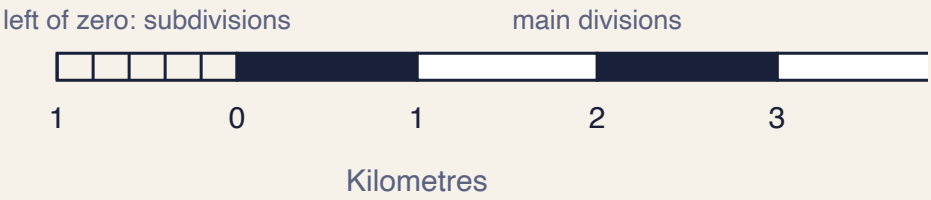
SIDEKICK TIP

Think of a staircase of importance: metalled road, unmetalled road, cart track, pack track, footpath. Each is one step narrower than the one before it.

## 4. Scale, Distance and Area

Scale links the map to the ground. There are three ways to show it.

| TYPE | HOW IT LOOKS | EXAMPLE |
|---|---|---|
| Statement scale | Words | 1 cm on the map represents 500 m on the ground |
| Representative Fraction (RF) | A ratio with no units | 1 : 50,000 (1 unit on map = 50,000 of the same unit on land) |
| Linear (graphical) scale | A ruler printed on the sheet | See below; read it with a strip of paper |




A linear scale. The part left of zero is divided into smaller units for fine readings.

**1. Straight lines**
Measure the map distance in cm with a ruler (or lay a strip of paper and compare with the linear scale). Multiply by the ground value of 1 cm.

**2. Curved routes**
Lay a thread along the road or river, straighten it and measure. The syllabus asks for direct distance (straight line) only.

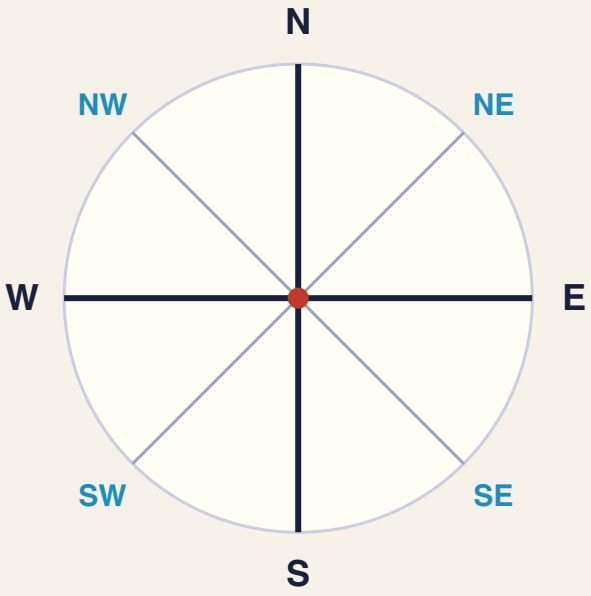
**3. Area**
Count the
**full grid squares**
inside the boundary. At 1:50,000 each is 1 sq km, so 12 full squares is 12 sq km. Ignore part squares.

**4. Conversion**
1:50,000 means 1 cm = 50,000 cm = 500 m, so 2 cm = 1 km and 2 cm x 2 cm = 1 sq km.

BOARD ALERT
Do not forget units, and do not mix cm and km in one step. Write the working: **3.4 cm x 0.5 km = 1.7 km**.

## 5. Direction

The top of the sheet is north. Use the eight cardinal points to describe where one place lies from another, always stating from which place: *'the temple lies north-east of the school'*. Draw a light pencil line between the two points and compare with the compass.

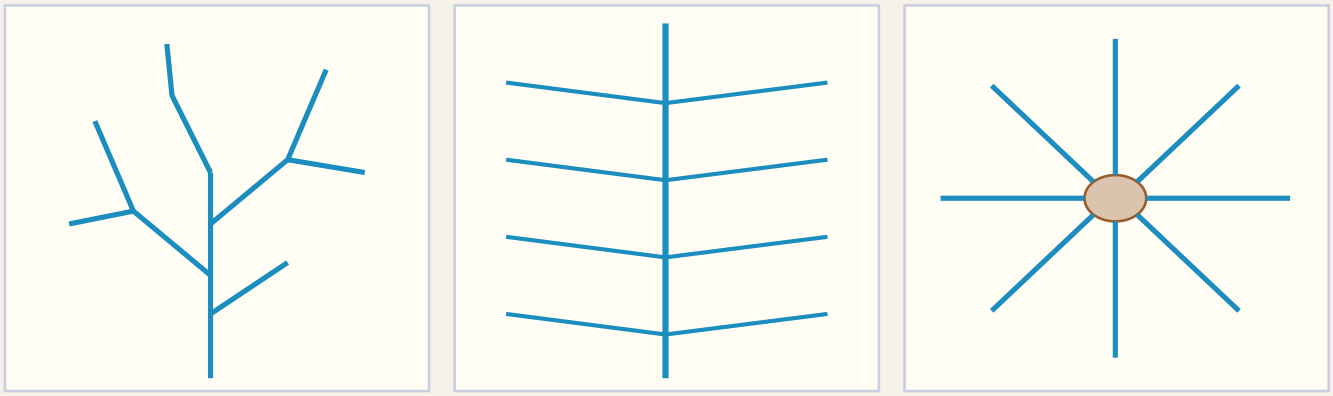



Eight cardinal points: N, NE, E, SE, S, SW, W, NW.

## 6. Drainage Patterns and Direction of Flow

Rivers and streams are drawn in blue. Smaller streams that join a bigger one are **tributaries**; the point where they meet is a **confluence**. The river flows **from higher to lower contour values**, and tributaries join at an angle that points downstream. Seasonal streams are drawn as broken blue lines, perennial ones as solid lines.

| PATTERN | LOOK | WHERE YOU FIND IT |
|---|---|---|
| Dendritic | Branches like a tree | Uniform rock on gentle slopes, e.g. the northern plains |
| Trellised | Main river with tributaries joining at right angles | Parallel ridges and valleys of folded land |
| Radial | Streams spread outward from a central height | Domed hills or volcanic cones |



**Dendritic (tree-like)** **Trellised (rectangular)** **Radial (from a hill)**

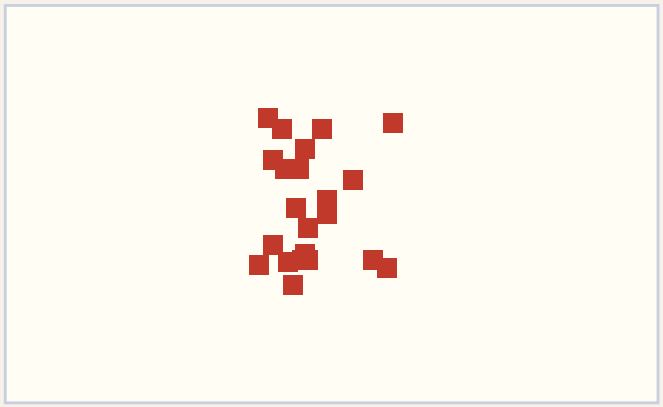
Blue lines are streams. The pattern depends on the slope and the rocks beneath.

SIDEKICK TIP

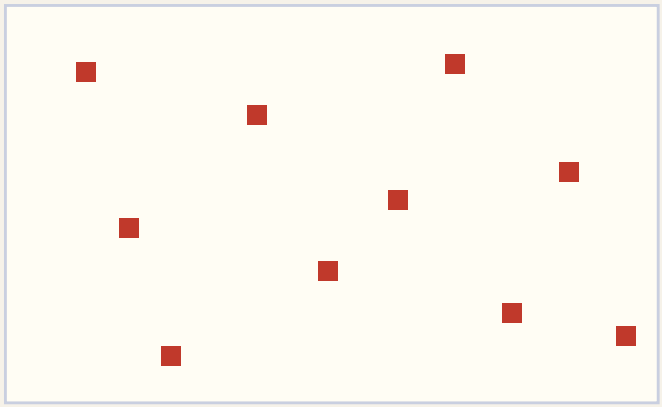
To say which way a river flows, find two contour values it crosses. The river runs from the larger number to the smaller. Left bank and right bank are named looking **downstream**.

## 7. Settlement Patterns, Site and Land Use

| PATTERN | DESCRIPTION | FAVOURED BY |
|---|---|---|
| Clustered (nucleated) | Houses packed close together | Fertile flat land, water, crossroads |
| Dispersed | Isolated homes or hamlets far apart | Hills, forest, dry land, thin soil |



**Clustered (nucleated)**



**Dispersed (scattered)**

Each red square is a house. Clusters form around water and flat land; dispersed homes suit rough or dry land.

The **site** of a village is the exact ground it stands on; the **situation** describes where it lies in relation to the surroundings. Good sites are near water, on slightly raised dry ground, on flat or gently sloping land, close to roads and farmland.

Land use is read from colour: yellow is crop land, green is forest or orchards, white is waste land, blue is water, and brown dots show sand. Occupations follow: farming where there are fields, trade and services at road junctions and railway stations, mining or quarrying where the key shows a mine.

### Natural and Man-made Features

| NATURAL | MAN-MADE |
|---|---|
| Rivers, streams, lakes, hills, ridges, valleys, forests, sand dunes | Roads, railways, canals, tanks, wells, bridges, villages, towns, temples, schools, post offices, boundaries |

REAL LIFE

The Survey of India sheet of your own town is on sale at its offices. Read it with this chapter and you will remember every symbol for life.

HOT QUESTION

A river in a toposheet shows a V-shaped bend in the 140 m contour pointing north-east, and the next contour to the south-west is 120 m. In which direction does the river flow, and what does the spacing of the contours tell you about the land?

## Mind Map Recap

### Topographical Maps

**Grid reference**
- Eastings first, northings second
- 4-figure = square; 6-figure = point

**Relief**
- Contour = line of equal height
- Close = steep, wide = gentle
- Hill, ridge, escarpment, valley

**Colours**
- Blue water, brown relief, green vegetation
- Yellow crops, red settlements, black man-made

**Scale**
- Statement, RF, linear
- Distance = map length x scale
- Area = full grid squares

**Direction**
- 8 cardinal points
- Always say 'from which place'

**Patterns**
- Dendritic, trellised, radial
- Clustered and dispersed
- Natural vs man-made

## Practice Questions

**Q1. Define contour and contour interval.**
*Answer:* A contour is a line on a map joining places of equal height above mean sea level. The contour interval is the constant height difference between two successive contours.

**Q2. How is a six-figure reference different from a four-figure one?**
*Answer:* A four-figure reference names a whole 1 km square by easting then northing. A six-figure reference adds a tenth-of-square digit to each, pointing to a spot about 100 m across.

**Q3. On a 1:50,000 map, two villages are 7.2 cm apart. Find the ground distance.**
*Answer:* 7.2 cm x 0.5 km per cm = 3.6 km.

**Q4. The area of a lake covers 9 full grid squares on a 1:50,000 sheet. What is its area?**
*Answer:* Each full square is 1 km x 1 km = 1 sq km, so the area is 9 sq km.

**Q5. What does it mean when contours are very close on one side of a hill and widely spaced on the other?**
*Answer:* The hill has a steep slope on the first side and a gentle slope on the other. A landform with such a profile is an escarpment.

**Q6. Name the colours used for: (a) water (b) relief (c) vegetation.**
*Answer:* Blue, brown and green, respectively.

**Q7. Give two differences between a spot height and a triangulated height.**
*Answer:* A spot height is shown by a dot and is levelled on the ground; a triangulated height is shown by a triangle and is fixed by trigonometrical survey, so it is more precise.

**Q8. Which drainage pattern forms when streams spread outward from a central hill?**
*Answer:* Radial pattern.

**Q9. Why do villages often cluster along a river bank in the plains?**
*Answer:* Flat fertile land and a dependable water supply attract people and farming, so houses gather close together (a clustered pattern).

**Q10. State two ways in which a dispersed settlement differs from a clustered settlement.**
*Answer:* Houses are scattered far apart rather than packed together, and are found on hills, dry or forested land rather than fertile plains.

02

CHAPTER · GEOGRAPHY

# Map of India: Mark and Identify

Ten marks are decided by what you can place on an outline map. This chapter gives you the exact checklist, a way to remember each position, and the habits that keep a map answer clean.



IN THIS CHAPTER

- 6 features to mark
- 4 features to identify
- Mnemonics and tips
- Quick-look sketch maps

## How the Map Question Works

The paper gives you an outline map of India. You are asked to **mark and name 6 features** and **identify 4 features** (usually numbered points, lines or shaded areas that you name). The two lists below are the complete syllabus.

## Features to MARK and NAME

| GROUP | MARK THESE |
|---|---|
| Mountains and peaks | Himalayas, Karakoram, Aravali, Vindhyas, Satpura, Western Ghats, Eastern Ghats, Nilgiris, Khasi Hills, Mount Godwin Austen (K2), Mount Kanchenjunga |
| Plateaus | Deccan Plateau, Chota Nagpur Plateau |
| Plains | Gangetic Plains; Coastal plains: Konkan, Malabar, Coromandel, Northern Circars |
| Desert | Thar (Great Indian Desert) |
| Water bodies | Gulf of Kutch, Gulf of Khambhat, Gulf of Mannar, Palk Strait, Andaman Sea, Wular Lake |
| Pass | Karakoram Pass |
| Winds | South-West Monsoon (Arabian Sea branch and Bay of Bengal branch), North-East Monsoon, Western Disturbances |
| Soil | Alluvial, Black, Red |
| Population | Dense and sparse distribution |

## Features to IDENTIFY

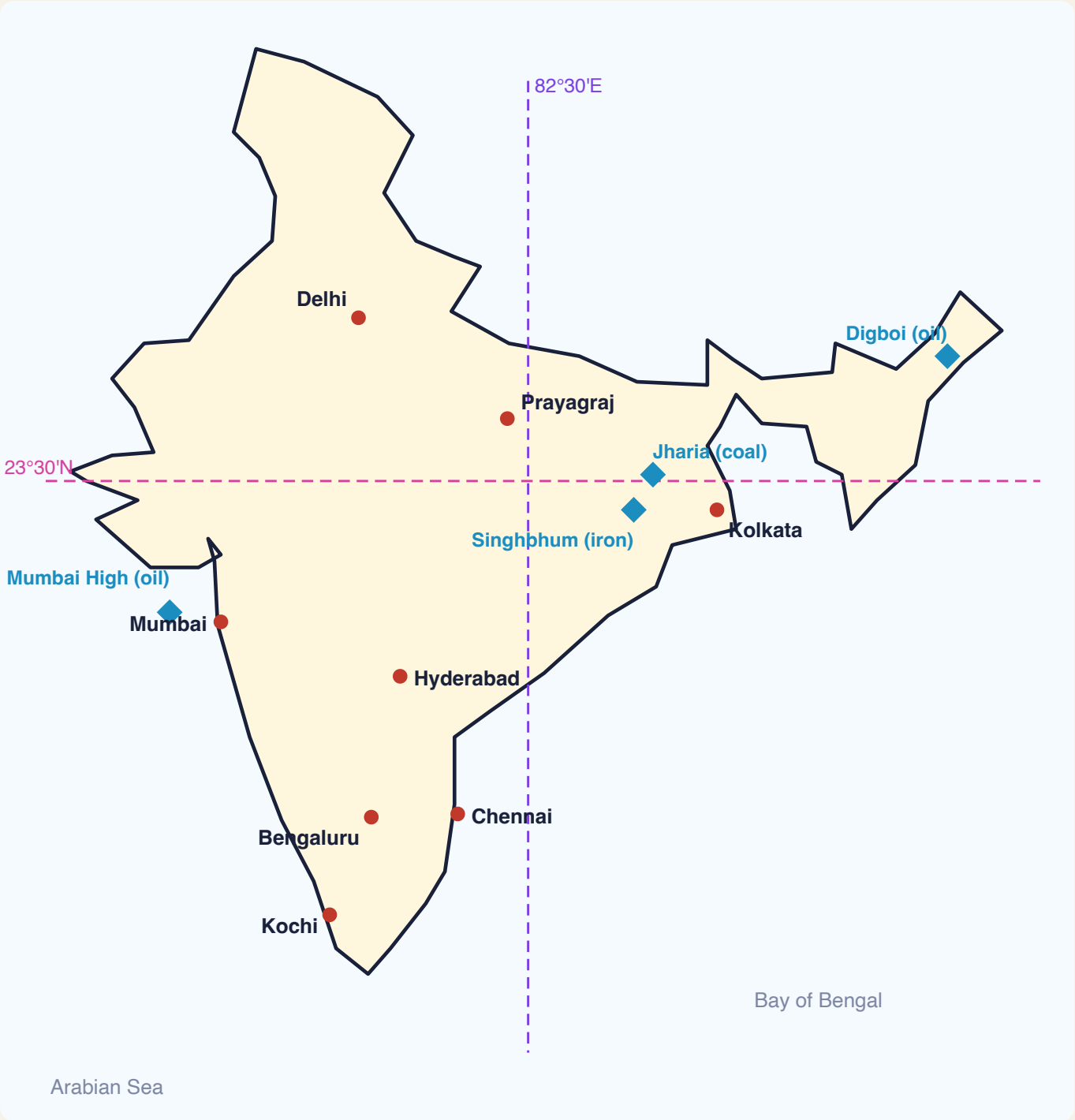
| GROUP | IDENTIFY THESE |
|---|---|
| Rivers | Indus, Ravi, Beas, Chenab, Jhelum, Satluj, Ganga, Yamuna, Ghaghra, Gomti, Gandak, Kosi, Chambal, Betwa, Brahmaputra, Narmada, Tapti, Mahanadi, Godavari, Krishna, Cauvery |
| Latitude and longitude | Tropic of Cancer, Standard Meridian (82°30'E) |
| Minerals | Oil: Mumbai High (offshore) and Digboi. Iron ore: Singhbhum. Coal: Jharia |
| Cities | Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bengaluru, Kochi, Allahabad (Prayagraj) |



KEY POINT

Mark = you place and label it on a blank map. Identify = you recognise something already drawn and name it.

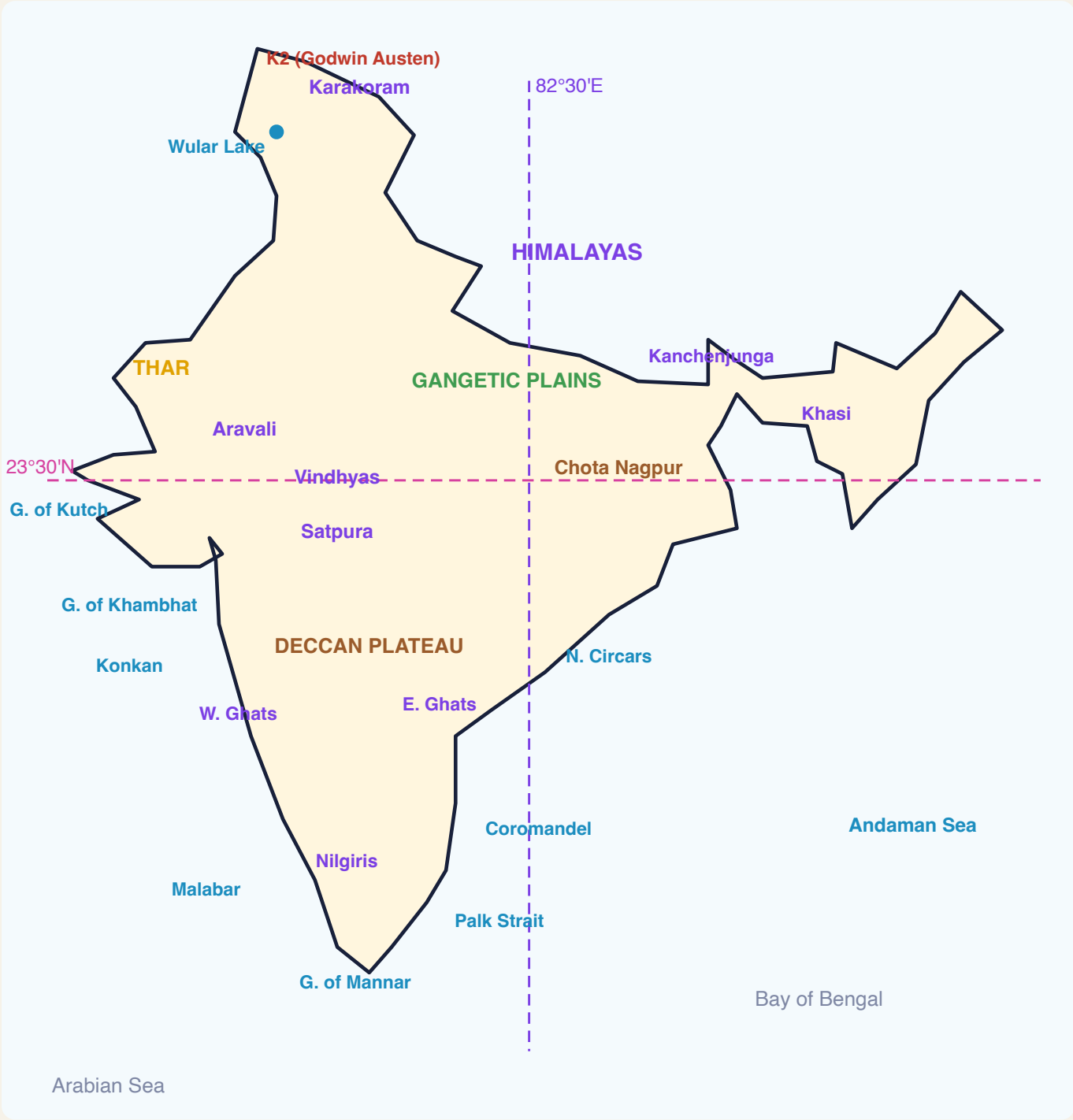
## Cities and Minerals




Schematic map (not to scale). Red dots: cities. Blue diamonds: minerals. Pink dashed line: Tropic of Cancer (23°30'N). Violet dashed line: Standard Meridian.

- The **Tropic of Cancer** (23°30'N) passes through Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.
- The **Standard Meridian** (82°30'E) passes near Mirzapur in Uttar Pradesh, a little east of Prayagraj. India's time is set by it, 5 hours 30 minutes ahead of GMT.
- **Jharia** (Jharkhand) is the coal capital, **Singhbhum** (Jharkhand) holds iron ore, **Digboi** (Assam) is the oldest oil field and **Mumbai High** lies offshore in the Arabian Sea.

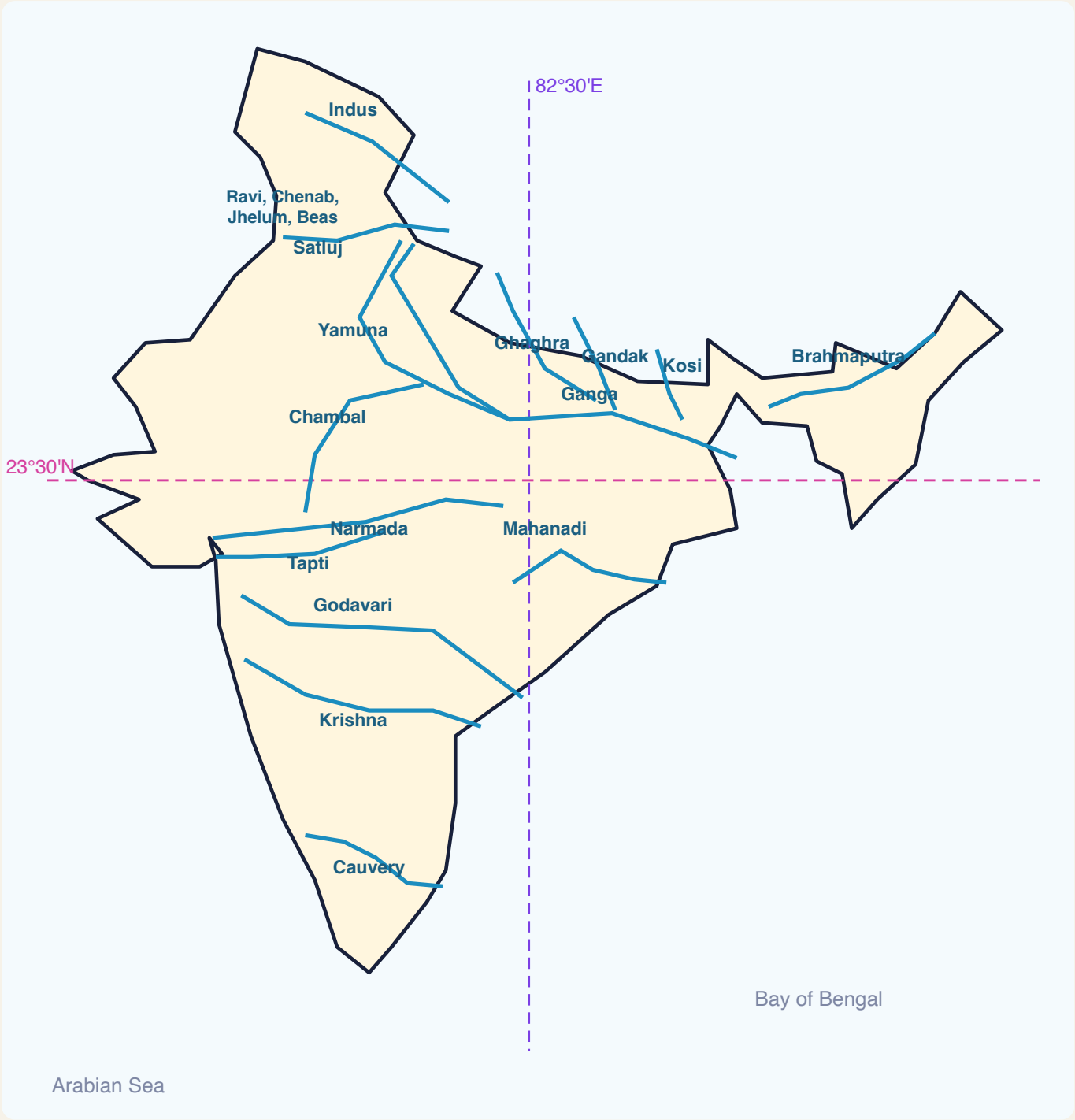
# Mountains, Plateaus, Plains and Water Bodies




Schematic relief map for revision. Positions are approximate. Mark these on the exam outline map in the same way.

| FEATURE | WHERE TO PLACE IT |
|---|---|
| Himalayas | Northern arc from Jammu and Kashmir to Arunachal Pradesh |
| Karakoram | North of Ladakh, west of the Himalayas; K2 (Godwin Austen, 8,611 m) lies here |
| Kanchenjunga | On the Sikkim-Nepal border (8,586 m), the highest peak in India |
| Aravali | A north-east to south-west line from Delhi to Gujarat through Rajasthan |
| Vindhyas, Satpura | East-west ranges north and south of the Narmada in central India; Vindhyas are the northern ones |
| Western Ghats / Nilgiris | Along the west coast; the Nilgiris are where the two Ghats meet in Tamil Nadu |
| Eastern Ghats | Broken hills along the east coast, Odisha to Tamil Nadu |
| Khasi Hills | Meghalaya, in the north-east |
| Deccan / Chota Nagpur | Deccan is the large triangle south of the Narmada; Chota Nagpur is its north-eastern part in Jharkhand |
| Wular Lake | Kashmir valley, on the Jhelum |

# Rivers (Identify)




Schematic river map. Gomti and Betwa are minor tributaries (of the Ganga and Yamuna) that you must be able to identify in a list.

| RIVER | SOURCE (REGION) | FLOWS INTO | KEY FACT |
|---|---|---|---|
| Indus | Near Lake Mansarovar (Tibet) | Arabian Sea (in Pakistan) | Enters India in Ladakh |
| Satluj | Tibet near Mansarovar | Beas, then Chenab (in Pakistan) | Bhakra dam |
| Ganga | Gangotri glacier (as Bhagirathi) | Bay of Bengal | Receives Yamuna, Ghaghra, Gomti, Gandak, Kosi |
| Yamuna | Yamunotri glacier | Ganga at Prayagraj | Delhi and Agra lie on it |
| Brahmaputra | Tibet (as Tsangpo) | Bay of Bengal (with the Ganga) | Called Siang, then Brahmaputra, in Assam |
| Narmada, Tapti | Central India | Arabian Sea | West-flowing; rift valleys |
| Mahanadi | Chhattisgarh | Bay of Bengal | Hirakud dam |
| Godavari | Near Nashik (Maharashtra) | Bay of Bengal | Longest peninsular river |
| Krishna | Mahabaleshwar (Maharashtra) | Bay of Bengal | Third longest peninsular river |
| Cauvery | Brahmagiri hills, Karnataka | Bay of Bengal | Shivasamudram falls |
| Chambal | Madhya Pradesh | Yamuna | Ravines (badlands) |



SIDEKICK TIP

Himalayan rivers: Indus system on the left-hand north-west (Indus, Jhelum, Chenab, Ravi, Beas, Satluj), Ganga system in the middle (Yamuna, Ghaghra, Gomti, Gandak, Kosi) and Brahmaputra in the far east. Peninsular rivers: Narmada and Tapti flow west, the rest flow east.

## Winds, Soils and Population

| ITEM | HOW TO MARK IT |
|---|---|
| South-West Monsoon | Two arrows: Arabian Sea branch hits the west coast; Bay of Bengal branch curves north-west up the Ganga plain |
| North-East Monsoon | An arrow from the north-east over land toward the Bay of Bengal, ending on the Tamil Nadu coast |
| Western Disturbances | An arrow from the north-west into Punjab and the Himalayan foothills |
| Alluvial soil | Gangetic plain, Brahmaputra valley and coastal plains |
| Black soil | Deccan lava plateau: Maharashtra, Gujarat, Madhya Pradesh |
| Red soil | Eastern and southern Peninsula: Tamil Nadu, Karnataka, Odisha, Jharkhand |
| Dense population | Ganga plain, coastal plains, Kerala; Kolkata, Mumbai belts |
| Sparse population | Himalayas, Thar desert, north-east hills, Rann of Kutch |



BOARD ALERT

Never write a name over a dense part of the map so that the border is unreadable. Draw a short pointer line and write the label in the sea or margin.

## Examiner's Tips

- Mark with a **sharp pencil** and use **neat capital letters**; keep a small key for shading.
- Use **one symbol per feature**: triangle for peaks, dot for cities, line for passes, arrows for winds.
- Place a river by its **source and mouth**, not by the whole route.
- Practise by tracing the outline from memory five times; the shape fixes your sense of position.
- If a spelling is doubtful, write the full name from the list above. Do not shorten to initials.



HOT QUESTION

Name three cities that lie close to the Tropic of Cancer and three that lie on or close to the coast. Which of the eight listed cities is nearest to the Standard Meridian?

## Mind Map Recap

**Map of India**

**Mountains**

- Himalayas, Karakoram, K2, Kanchenjunga
- Aravali, Vindhyas, Satpura
- W. and E. Ghats, Nilgiris, Khasi

**Land**

- Deccan, Chota Nagpur
- Gangetic and coastal plains
- Thar desert

**Water**

- Gulfs, Palk Strait, Andaman Sea, Wular Lake
- 21 rivers to identify

**Lines**

- Tropic of Cancer 23°30'N
- Standard Meridian 82°30'E

**Resources**

- Mumbai High, Digboi
- Singhbhum, Jharia
- 8 cities

**Climate and people**

- Monsoon arrows
- 3 soils
- Population dense and sparse

## Practice Questions

**Q1. Which latitude passes through the centre of India, and through how many states does it pass?**
*Answer:* The Tropic of Cancer (23°30'N). It passes through eight states: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

**Q2. Name the two oil fields to identify on the map.**
*Answer:* Digboi in Assam and Mumbai High, an offshore field off the Maharashtra coast.

**Q3. Which water body separates India from Sri Lanka, and which gulf lies to its south-west?**
*Answer:* The Palk Strait separates them; the Gulf of Mannar lies to the south of it.

**Q4. Name the four coastal plains in the syllabus.**
*Answer:* Konkan, Malabar (west coast), Coromandel and Northern Circars (east coast).

**Q5. Where is Wular Lake?**
*Answer:* In the Kashmir valley (Jammu and Kashmir), on the Jhelum river.

**Q6. Which two ranges form the northern and southern sides of the Narmada valley?**
*Answer:* The Vindhyas on the north and the Satpura on the south.

**Q7. Name two Himalayan peaks that must be marked and the ranges they belong to.**
*Answer:* Mount Godwin Austen (K2) in the Karakoram, and Kanchenjunga in the eastern Himalayas on the Sikkim-Nepal border.

**Q8. Which river flows west into the Arabian Sea through a rift valley: Godavari or Narmada?**
*Answer:* The Narmada.

**Q9. Where would you mark the Western Disturbances on the map?**
*Answer:* As an arrow entering from the north-west, towards Punjab, Haryana and the Himalayan foothills.

**Q10. Why should names be written outside crowded areas with pointers?**
*Answer:* To keep the map clear and the names legible; examiners cannot give marks for a label that hides a border or another feature.

# 03

CHAPTER · GEOGRAPHY

# Location, Extent and Physical Features

India's position on the globe decides its climate, its trade routes and its neighbours. Its physical features decide where rivers flow, where soil is rich and where people live. Learn where things are and why they matter.



IN THIS CHAPTER

- Location and extent
- Neighbours and frontiers
- Himalayas
- Northern plains
- Peninsular plateau
- Rivers, coasts and islands

## Where Is India?

- India lies in the **northern and eastern hemispheres**, in the south of Asia.
- **Latitudes:** 8°4'N to 37°6'N (mainland south end at Kanyakumari). **Longitudes:** 68°7'E to 97°25'E.
- The **Tropic of Cancer** (23°30'N) cuts the country almost in half. The south is tropical; the north is subtropical or temperate.
- **Area:** about 3.28 million sq km, the **seventh largest country** in the world (about 2.4% of the land area).
- **North to south** distance is about 3,214 km and **east to west** is about 2,933 km.
- **Land frontier** about 15,200 km, **coastline** about 7,516 km including the islands.
- **Standard Meridian** 82°30'E, which gives Indian Standard Time (5 h 30 min ahead of GMT). Longitude difference between Gujarat and Arunachal is about 30°, that is about 2 hours of local time.

| NEIGHBOUR | DIRECTION | BORDER |
|---|---|---|
| Pakistan, Afghanistan | North-west | Land |
| China (Tibet), Nepal, Bhutan | North | Land (the Himalayas) |
| Myanmar | East | Land |
| Bangladesh | East | Land (longest border, touching five states: West Bengal, Assam, Meghalaya, Tripura and Mizoram) |
| Sri Lanka | South | Across the Palk Strait and Gulf of Mannar |



SIDEKICK TIP

India is a 'peninsula with a crown'. Water on three sides (Arabian Sea in the west, Bay of Bengal in the east, Indian Ocean in the south) and a wall of mountains on the fourth.



REAL LIFE

The Tropic of Cancer lies between Delhi and Mumbai in latitude. Delhi (28°N) is north of it; Kolkata (22°N) is just south of it.

## Why India's Location Matters

- The **Indian Ocean** sits in the middle of sea routes between West Asia, Africa and East Asia. Ports like Mumbai and Chennai gain from this.
- The **Himalayas** shield India from cold northern winds and bring monsoon rain.
- The **mountain passes of the north-west** linked India with Central Asia, which is why most invaders and traders came that way.
- A **central position on the Suez route** let India trade with Europe, Africa and the Far East for centuries.

## Physical Divisions

Geographers divide India into six regions by relief.

| DIVISION | WHERE | FEATURES |
|---|---|---|
| Northern Mountains | Northern edge, Kashmir to Arunachal | Young fold mountains, snow, glaciers, source of great rivers |
| Northern Plains | Between the Himalayas and the Peninsula | Alluvial, flat, fertile, densely populated |
| Peninsular Plateau | South of the plains | Ancient hard rocks, mineral rich |
| Indian Desert | Rajasthan, west of the Aravali | Sand dunes, scanty rainfall |
| Coastal Plains | Along both coasts | Narrow strips, ports, paddy fields |
| Islands | Bay of Bengal, Arabian Sea | Andaman and Nicobar; Lakshadweep |

## 1. The Himalayas

The Himalayas are young **fold mountains** formed by the Indian plate colliding with the Eurasian plate. They stretch about 2,400 km from the Indus gorge in the west to the Brahmaputra gorge in the east, and have three parallel ranges.

| RANGE | OTHER NAMES | HEIGHT AND FEATURES |
|---|---|---|
| Greater Himalaya | Himadri | Highest, permanently snow-covered, over 6,000 m; peaks such as Kanchenjunga, Nanda Devi |
| Lesser Himalaya | Himachal | 3,700 to 4,500 m; hill stations such as Shimla, Mussoorie, Nainital, Darjeeling |
| Outer Himalaya | Shiwaliks | 900 to 1,100 m; lowest and youngest; duns (valleys) such as Dehradun |

- **Karakoram** range in the north-west holds K2 or Mount Godwin Austen (8,611 m), second highest in the world, in the Karakoram (in territory India claims as part of Ladakh).
- **Kanchenjunga** (8,586 m) is the third highest peak in the world and the highest peak in India (on the Sikkim-Nepal border). **Nanda Devi** (7,816 m) is the highest peak lying wholly in India.
- Important **passes**: Karakoram, Zoji La (Kashmir to Ladakh), Banihal, Shipki La, Nathu La, Bara Lacha La.
- The **eastern hills** (Purvanchal): Patkai, Naga, Mizo and the Khasi, Garo and Jaintia hills of Meghalaya.



FUN FACT

The Himalayas are still rising by about 5 mm a year, which is why the region has frequent earthquakes.

## 2. The Northern Plains

Built by silt deposited by the **Indus, Ganga and Brahmaputra** over millions of years. They are about 2,400 km long and 240 to 320 km wide, and stretch from Punjab to Assam.

| PART | REGION | RIVER SYSTEM |
|---|---|---|
| Punjab plain | Punjab, Haryana | Indus and tributaries (Beas, Satluj, Ravi) |
| Ganga plain | Uttar Pradesh, Bihar, West Bengal | Ganga and tributaries |
| Brahmaputra plain | Assam | Brahmaputra |

- **Bhangar** is the older, higher alluvium with kankar; **khadar** is the newer, lower, more fertile alluvium that floods yearly.
- Fertile soil, plenty of water and flat land make this the **food bowl** of India.

## 3. The Peninsular Plateau

A tableland of very old crystalline and volcanic rocks. It is rich in minerals. It has two parts: the **Central Highlands** (Malwa plateau, Chota Nagpur plateau, Bundelkhand) north of the Narmada, and the **Deccan Plateau** south of it, tilting slightly eastward.

| HILLS | WHERE | NOTES |
|---|---|---|
| Aravali | Delhi to Gujarat via Rajasthan | Among the oldest fold mountains in the world; Guru Shikhar (Mt Abu, 1,722 m) is the highest |
| Vindhya | Central India | North of the Narmada, east-west |
| Satpura | Central India | South of the Narmada; Dhupgarh (1,350 m) is the highest |
| Western Ghats | Along the west coast | Continuous; steep to the west; Anai Mudi (2,695 m) is the highest in the south |
| Eastern Ghats | Along the east coast | Broken by rivers; Mahendragiri (1,501 m) |
| Nilgiris | Where the Ghats meet | Doddabetta (2,637 m) |



KEY POINT

The Narmada and Tapti flow **west** through rift valleys. All the other major peninsular rivers flow **east** to the Bay of Bengal.

## 4. The Indian Desert

The **Thar** lies west of the Aravali. It receives under 25 cm of rain, has shifting sand dunes (**barchans**) and a seasonal river, the **Luni**. Camels are the typical transport.

## 5. Coastal Plains and Islands

| PLAIN | LOCATION | FEATURES |
|---|---|---|
| Konkan | West coast, Mumbai to Goa | Narrow, rocky, with many creeks |
| Malabar | Kerala coast | Backwaters, lagoons, coconut |
| Coromandel | East coast, southern Andhra Pradesh to Tamil Nadu | Broad, receives winter rain from retreating monsoon |
| Northern Circars | East coast, Andhra Pradesh and Odisha, between Mahanadi and Krishna deltas | Deltas of the Godavari and Krishna |

India has two island groups: **Andaman and Nicobar** in the Bay of Bengal (about 570 islands, highest point Saddle Peak) and **Lakshadweep** in the Arabian Sea (small coral atolls).

## Drainage: The River Systems

| | HIMALAYAN RIVERS | PENINSULAR RIVERS |
|---|---|---|
| Source | Glaciers and snow | Plateau, rainfall |
| Flow | All year (perennial) | Seasonal, depend on rain |
| Valley | Deep V-shaped gorges, delta at mouth | Shallow valleys, many with deltas or estuaries |
| Examples | Indus, Ganga, Brahmaputra | Godavari, Krishna, Cauvery, Mahanadi, Narmada, Tapti |
| Use | Irrigation, hydropower, navigation | Irrigation and hydropower with dams |

- **Ganga** rises at Gangotri (as the Bhagirathi), becomes the Ganga at Devprayag and flows through UP, Bihar, West Bengal. The **Yamuna** joins at Prayagraj. In Bangladesh it joins the Brahmaputra.
- **Brahmaputra** flows through Tibet (as Tsangpo), Arunachal (as Siang) and Assam. It floods every year.
- **Indus** rises in Tibet and enters India in Ladakh. Its Indian tributaries are the Jhelum, Chenab, Ravi, Beas and Satluj.
- **Godavari** is the longest peninsular river (Dakshin Ganga).
- **Lakes:** Wular (largest freshwater, Kashmir), Dal (Srinagar), Chilika (brackish lagoon, Odisha), Sambhar (salt, Rajasthan), Pulicat (lagoon on Andhra-Tamil Nadu border).

BOARD ALERT

Do not say all peninsular rivers flow east. Narmada and Tapti drain into the Arabian Sea. Do not call the Brahmaputra 'Ganga'; they meet only in Bangladesh.

HOT QUESTION

The Himalayan rivers are perennial but the peninsular rivers are seasonal. Give reasons, and explain why the plains of the Ganga are more densely populated than the plateau.

# Mind Map Recap

## Physical Features

### Location
- 8°4'N-37°6'N; 68°7'E-97°25'E
- Tropic of Cancer; 82°30'E
- 7th largest, ~3.28 million sq km

### Himalayas
- Himadri, Himachal, Shiwaliks
- K2, Kanchenjunga, Nanda Devi

### Plains
- Indus, Ganga, Brahmaputra
- Bhangar and khadar

### Plateau
- Aravali, Vindhya, Satpura
- Deccan, Chota Nagpur
- W. and E. Ghats

### Coast
- Konkan, Malabar
- Coromandel, Northern Circars
- Andaman, Lakshadweep

### Rivers
- Himalayan: perennial
- Peninsular: seasonal
- Narmada, Tapti west-flowing

## Practice Questions

**Q1. Why is the Tropic of Cancer important for India?**
*Answer:* It passes almost through the middle of the country, dividing it into a tropical south and a subtropical north, and it passes through eight states.

**Q2. Name the three parallel ranges of the Himalayas, from north to south.**
*Answer:* Himadri (Greater), Himachal (Lesser) and Shiwaliks (Outer).

**Q3. Differentiate between bhangar and khadar.**
*Answer:* Bhangar is old alluvium, higher and often with kankar; khadar is new alluvium, lower and more fertile, renewed by floods.

**Q4. Why do the Narmada and Tapti flow west?**
*Answer:* They flow in rift valleys (fault troughs) that slope westward, so the rivers have no choice but to run to the Arabian Sea.

**Q5. What is the Standard Meridian of India and why was it chosen?**
*Answer:* 82°30'E; it runs through the middle of the country and gives one uniform time (IST, GMT+5:30) for all places.

**Q6. Compare the Western and Eastern Ghats in two points.**
*Answer:* The Western Ghats are higher and continuous, running steeply along the coast; the Eastern Ghats are lower and broken by river valleys.

**Q7. Which is the highest peak entirely within India?**
*Answer:* Nanda Devi (7,816 m) in Uttarakhand.

**Q8. Name two islands groups of India and the water bodies they lie in.**
*Answer:* Andaman and Nicobar in the Bay of Bengal; Lakshadweep in the Arabian Sea.

**Q9. State two differences between Himalayan and peninsular rivers.**
*Answer:* Himalayan rivers are perennial and form gorges and large deltas; peninsular rivers are seasonal and flow in shallower valleys.

**Q10. Why are the northern plains densely populated?**
*Answer:* Flat land, fertile alluvial soil, perennial rivers, good transport and a mild climate support farming and settlement.

04

CHAPTER · GEOGRAPHY

# Climate of India

India has a monsoon climate. Winds reverse with the seasons, farmers wait for the first rain, and the whole economy rides on the monsoon. See how temperature, pressure and wind combine.



IN THIS CHAPTER

- Factors of climate
- Monsoon mechanism
- Four seasons
- Local winds
- Rainfall zones
- Climate data

## Weather or Climate?

**Weather** is the state of the atmosphere at a place on a given day. **Climate** is the average of that weather over 30 years or more. India's climate is **tropical monsoon**: hot and wet summers, dry winters, with huge variation between regions.

## Factors That Shape India's Climate

| FACTOR | HOW IT AFFECTS INDIA |
|---|---|
| Latitude | The Tropic of Cancer divides the country. The south is hot all year; the north has a wide range of temperature |
| Altitude | Temperature falls about 6.5°C for every 1,000 m. Shimla, Ooty and Darjeeling are cool in summer |
| Distance from the sea | Coasts have moderate (equable) climate with high humidity, e.g. Mumbai. The interior has extremes, e.g. Delhi, Jaipur |
| Monsoon winds | Seasonal reversal of winds brings India most of its rain |
| Himalayas | A barrier that stops cold Central Asian winds and forces monsoon winds to rise and shed rain |
| Jet streams and El Nino | Upper atmospheric winds and Pacific Ocean warming that control the arrival and strength of monsoon (see the table below) |

| TERM | MEANING | EFFECT ON INDIA |
|---|---|---|
| Jet streams | Narrow, very fast bands of westerly winds in the upper atmosphere | In winter the subtropical westerly jet flows south of the Himalayas and steers western disturbances into India. In summer it shifts north of the Himalayas, helping the monsoon to burst over the country |
| Western disturbances | Weather systems from the Mediterranean region that travel east | Bring winter rain and snow to north-west India, good for wheat and the rabi crop |
| El Nino | Periodic warming of the eastern Pacific near Peru | Weakens the south-west monsoon, causing deficient rain and drought in many parts |



FUN FACT

El Nino is Spanish for 'the little boy' (the Christ child), named because the warm current often showed up around Christmas off Peru.

## The Monsoon: How It Works

The word comes from the Arabic *mausim*, meaning season. A monsoon is a system of **seasonal wind reversal**.

**1. 1. Land heats faster than sea**
In summer the north-west plains and the Thar heat to over 45°C. Air rises and a
**low pressure**
trough forms. The sea stays cooler, so pressure there is higher.

**2. 2. Winds cross the Equator**
The south-east trade winds of the southern hemisphere move toward the low pressure, cross the equator and are deflected to the right (Coriolis effect) to blow from the
**south-west**
.

**3. 3. Jet stream shifts**
Heating of Tibet and the north shift of the jet stream pulls the Inter-Tropical Convergence Zone (ITCZ) north. The monsoon
**bursts**
on Kerala in the first week of June.

**4. 4. Rain bearing winds split**
The Arabian Sea branch hits the Western Ghats and gives heavy rain on the windward side. The Bay of Bengal branch strikes the north-east hills and turns west up the Ganga plain.

In winter the process reverses. High pressure builds over the north-west of India, and cold dry air blows **from land to sea** as the **North-East monsoon**. It picks up moisture over the Bay of Bengal and gives rain on the Tamil Nadu coast in October to December.



KEY POINT
Cause of the Indian monsoon = **differential heating** of land and sea + the shift of the ITCZ + the position of the jet stream and the Tibetan plateau heating.

## The Four Seasons

| SEASON | MONTHS | TEMPERATURE AND PRESSURE | WINDS AND WEATHER |
|---|---|---|---|
| Summer (hot weather) | March to May | Temperature rises; 40-45°C in the north-west. Low pressure over the north-west | Dry hot **loo** in the plains; dust storms; local **mango showers** in the south; **kalbaisakhi (nor'westers)** in Bengal and Assam |
| Monsoon (rainy) | June to September | Humid and warm; falls with the rain. Low pressure trough over the Ganga plain | South-west monsoon brings about 75% of the year's rain. **Burst** in Kerala around 1 June, whole country by mid-July |
| Retreating monsoon | October to November | Clear skies, humid heat ('October heat'), then temperature falls. Pressure rises in the north | Winds turn to the north-east. Cyclones form in the Bay of Bengal and hit the east coast |
| Winter (cold weather) | December to February | Cold in the north (below 0°C in Kashmir), mild in the south (above 20°C). High pressure over the north-west | Dry north-east winds; **western disturbances** give winter showers in Punjab; Tamil Nadu coast gets rain |



SIDEKICK TIP
Season order: **S**ummer, **M**onsoon, **R**etreating, **W**inter. Remember: 'Sun Makes Rain Wet'.

## Local Winds and Showers

| NAME | WHERE | WHAT IT DOES |
|---|---|---|
| Loo | Northern plains, May-June | Hot dry wind; heat stroke risk |
| Mango showers | Kerala and Karnataka, April-May | Pre-monsoon rain that helps mangoes ripen |
| Kalbaisakhi (nor'wester) | West Bengal, Assam, Bangladesh | Violent thunderstorms with squalls; help tea, jute and rice |
| Blossom showers | Coffee belt of Karnataka and Kerala | Cause coffee flowers to bloom |
| Western disturbances | North-west India in winter | Rain and snow; good for wheat |

## Distribution of Temperature and Rainfall

- **Hottest places:** western Rajasthan and the north-west plains (summer 45 to 50°C). **Coldest:** Kashmir and Ladakh (Dras, Kargil, Leh).
- Southern coasts have an **annual range of only 3-5°C**; Delhi and Jaipur have more than 20°C.
- Average annual rainfall in India is about **118 cm**. **Mawsynram** in Meghalaya is among the wettest places on Earth. Jaisalmer in the Thar is among the driest.

| RAINFALL ZONE | ANNUAL RAINFALL | REGIONS | CROPS/VEGETATION |
|---|---|---|---|
| Heavy | Above 200 cm | Western Ghats, north-east India, Meghalaya hills, Andaman | Rainforest, rice, tea, rubber |
| Moderate | 100 to 200 cm | Eastern Maharashtra, MP, Odisha, Bihar, Lower Ganga plain | Rice, sugarcane, jute, deciduous forest |
| Low | 50 to 100 cm | Deccan interior, UP, Punjab, Gujarat | Wheat, millets, pulses, cotton |
| Scanty | Below 50 cm | West Rajasthan, Kutch, Ladakh | Thorn scrub, desert vegetation |

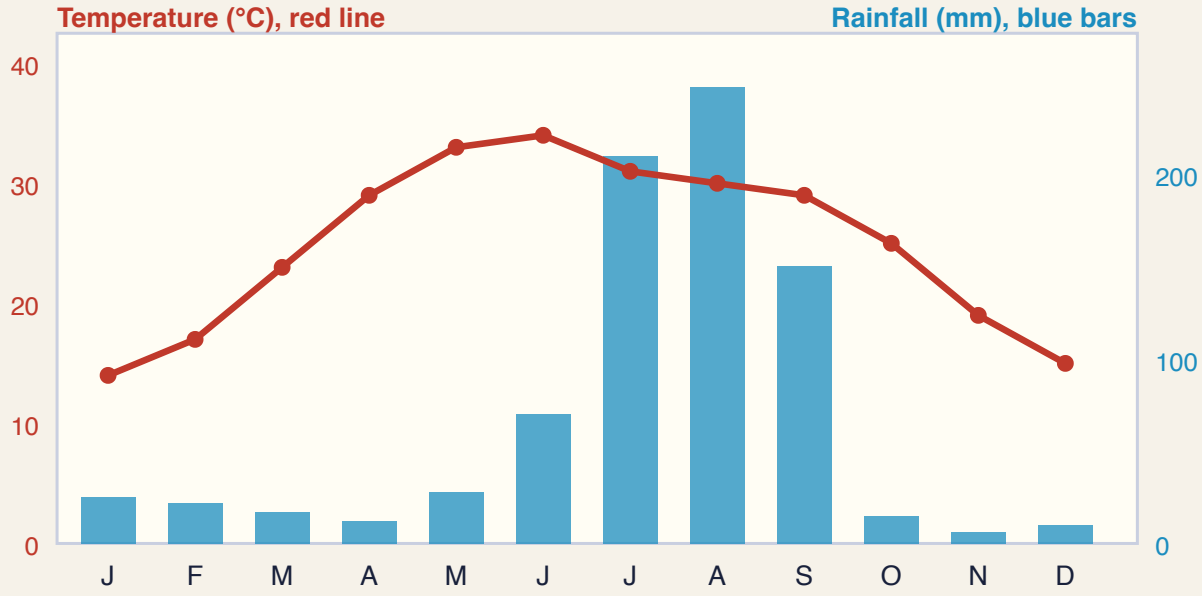
# Interpreting Climate Data

**1. Find the annual range**
Highest monthly mean temperature minus lowest. A small range means a coastal (equable) climate; a large range means an interior (continental) one.

**2. Total the rainfall**
Add all monthly rainfall values.

**3. Spot the rainy season**
The wettest months. June to September means the south-west monsoon.

**4. Name and justify**
Name the climate and link it to location: coast, interior, hill or desert.




A climograph for 'Station A' (a made-up northern plains station, used in the worked example).

| STATION A | J | F | M | A | M | J | J | A | S | O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Temp (°C) | 14 | 17 | 23 | 29 | 33 | 34 | 31 | 30 | 29 | 25 | 19 | 15 |
| Rain (mm) | 25 | 22 | 17 | 12 | 28 | 70 | 210 | 247 | 150 | 15 | 6 | 10 |

**Worked reading:** annual range = 34 - 14 = **20°C**; total rainfall = **812 mm**; wettest months July and August, so monsoon rain from the south-west; large range and moderate rain suggest an **interior north Indian plain** such as Delhi.

BOARD ALERT
Do not call 20°C a 'high temperature'. Range is a difference, not a reading. Write the unit every time.

HOT QUESTION
Why does Chennai get most rain in November and December while Mumbai gets it in July and August? Answer using the direction of the winds and the coast each faces.

# Mind Map Recap

**Climate**

### Factors
- Latitude, altitude, sea distance
- Monsoon winds, Himalayas
- Jet streams and El Nino

### Monsoon
- Land heats, low pressure
- SE trades turn SW
- Burst on Kerala ~1 June
- Winter: NE monsoon

### Seasons
- Summer Mar-May
- Monsoon Jun-Sep
- Retreating Oct-Nov
- Winter Dec-Feb

### Local
- Loo, mango showers
- Kalbaisakhi, blossom showers
- Western disturbances

### Rainfall
- Heavy over 200 cm
- Moderate 100-200
- Low 50-100
- Scanty under 50

### Data
- Annual range
- Total rainfall
- Rainy season
- Name the climate

## Practice Questions

**Q1. Define monsoon.**
*Answer:* A system of winds that reverse their direction with the season. In summer they blow from the sea to the land, in winter from the land to the sea.

**Q2. State two reasons why Shimla is cooler than Delhi in summer.**
*Answer:* It is at a higher altitude (temperature falls about 6.5°C per 1,000 m) and the surrounding hills and forests moderate the heat.

**Q3. What is meant by the 'burst' of the monsoon?**
*Answer:* The sudden arrival of the south-west monsoon with heavy rain and thunder, usually on the Kerala coast around the first week of June.

**Q4. Why do Coromandel coast districts get rain in winter?**
*Answer:* North-east winds blowing over the Bay of Bengal pick up moisture and shed it on the Tamil Nadu coast.

**Q5. What are western disturbances and why are they useful?**
*Answer:* Weather systems from the Mediterranean that bring winter rain and snow to north-west India, helping the wheat crop.

**Q6. Name two effects of El Nino on India.**
*Answer:* It weakens the south-west monsoon, giving deficient rain and drought, which hurts kharif crops and raises food prices.

**Q7. A station has monthly temperatures of 31°C (highest) and 21°C (lowest). What is the range and what does it suggest?**
*Answer:* 10°C, a small range, suggesting a coastal or equable climate.

**Q8. Why is the Rajasthan desert rain-starved?**
*Answer:* The Aravali lies parallel to the Arabian Sea branch, so moisture-bearing winds pass without being forced to rise; the region is also distant from the Bay of Bengal.

**Q9. Give two points of difference between the Arabian Sea branch and the Bay of Bengal branch of the monsoon.**
*Answer:* The Arabian Sea branch is stronger and hits the Western Ghats first; the Bay of Bengal branch strikes the north-east and turns west along the Ganga plain, giving less rain as it moves west.

**Q10. What is 'loo'?**
*Answer:* A hot, dry, dusty wind blowing over north India in May-June.

05

CHAPTER · GEOGRAPHY

# Soil Resources

Soil is the thin skin of the earth that feeds us. It takes hundreds of years to build and only a few seasons to lose. Know India's four main soils and how to protect them.



IN THIS CHAPTER

- Soil formation
- Alluvial soil
- Black soil
- Red soil
- Laterite soil
- Erosion and conservation

## What Is Soil and How Does It Form?

**Soil** is the loose upper layer of the earth made of weathered rock particles, humus, water, air and living organisms. Five things decide what a soil is like: **parent rock, climate, relief, vegetation and time**.

- Rocks break down by weathering (heat and cold, water, wind, plants).
- Rivers carry the fine particles and spread them as **alluvium**, or the rock decays in place as **residual soil**.
- Decayed plants and animals add **humus**, which gives fertility.

## Types of Soil in India

| | ALLUVIAL | BLACK (REGUR) | RED | LATERITE |
|---|---|---|---|---|
| Formation | Silt deposited by rivers (Ganga, Indus, Brahmaputra) | Weathered volcanic basalt (Deccan lava) | Weathering of granite and gneiss (crystalline rocks) | Leaching in heavy rain and high heat |
| Colour | Light grey to ash | Black to dark grey | Red to yellow | Brick red |
| Texture | Sandy loam to clay | Clayey, sticky when wet | Sandy to loam, porous | Coarse, gravelly, porous |
| Minerals | Rich in potash and lime; poor in phosphorus and nitrogen | Rich in calcium, magnesium, iron, lime; poor in nitrogen, phosphorus and humus | Rich in iron oxide (gives red colour); poor in nitrogen, humus, lime | Rich in iron and aluminium; poor in nitrogen, potash, lime |
| Characteristics | Very fertile, easy to plough | Retains moisture; cracks in summer, **self-ploughing**; sticky in rain | Moderate fertility; needs fertilisers and irrigation | Low fertility; hardens into bricks on drying |
| Found in | Northern plains, coastal plains, deltas | Maharashtra, Gujarat, MP, parts of Karnataka and Andhra | Tamil Nadu, Karnataka, Andhra, Odisha, Jharkhand, Chhattisgarh | Western Ghats, Odisha, Assam, Kerala, Karnataka |
| Crops | Rice, wheat, sugarcane, jute, oilseeds | **Cotton**, sugarcane, soyabean, wheat, jowar | Millets, groundnut, pulses, potato, tobacco | Tea, coffee, rubber, cashew, coconut |



SIDEKICK TIP

Link each soil with a keyword: **Alluvial = river plains = wheat. Black = cotton = self-ploughing. Red = iron oxide = millets. Laterite = leaching = tea and coffee.**

## Alluvial Soil in Two Types

**Khadar** is the new, light, sandy and more fertile alluvium of the floodplains. **Bhangar** is the old, darker, clayey alluvium on higher land with kankar (lime nodules). Alluvial soil covers the largest area and supports the most people.

## Soil Erosion

**Soil erosion** is the removal of topsoil by running water, wind, ice or human action faster than nature can form it.

| AGENT | TYPE OF EROSION | WHERE |
|---|---|---|
| Running water | Sheet erosion (thin layer removed); gully erosion (deep channels); ravines | Chambal ravines, hill slopes |
| Wind | Deflation, dust storms, dunes | Thar desert, dry plains |
| Glaciers and waves | Scouring and cliff cutting | Himalayas, coasts |

### Causes of Soil Erosion

- **Deforestation** exposes the soil to rain and wind.
- **Overgrazing** removes the grass cover.
- **Faulty farming**: ploughing up and down the slope, monoculture, jhum (shifting) cultivation.
- **Heavy rain** on bare slopes, and **construction** and mining that strip the cover.

## Soil Conservation

Soil conservation means preventing erosion and restoring soil fertility. Why it matters: a thin layer lost can take hundreds of years to rebuild, and food security depends on it.

| METHOD | HOW IT WORKS |
|---|---|
| Afforestation | Tree roots bind soil, canopy softens rain |
| Contour ploughing | Furrows along the contours slow water on slopes |
| Terrace farming | Steps cut in hillsides reduce run-off (Himalayas, Western Ghats) |
| Strip cropping | Alternate strips of crops and grass |
| Shelter belts | Lines of trees to cut wind in deserts |
| Check dams | Small barriers in gullies trap silt |
| Crop rotation, mulching, manure | Restore nutrients and cover the soil |
| Controlled grazing | Stops animals stripping the ground |

KEY POINT

Soil erosion removes the **topsoil**, the most fertile layer. Conservation = **ATCS**: Afforestation, Terracing, Contour ploughing, Shelter belts.

FUN FACT

Black soil is called *regur* in the Deccan. It can hold water for months, yet its cracks in summer help it 'swallow' the top layer and turn itself over, so it is nearly self-ploughing.

HOT QUESTION

A farmer in a hilly region of Himachal Pradesh complains that every monsoon the topsoil of his field is washed away. Suggest three methods he could use and say why each works.

## Mind Map Recap

**Soil**

**Formation**
- Parent rock, climate, relief, vegetation, time

**Alluvial**
- Silt of rivers; khadar and bhangar
- Rice, wheat, sugarcane, jute

**Black**
- Basalt; regur; self-ploughing
- Cotton

**Red**
- Granite and gneiss; iron oxide
- Millets, groundnut, pulses

**Laterite**
- Leaching; tea, coffee, rubber, cashew

**Erosion**
- Water, wind; deforestation, overgrazing
- Terracing, contour ploughing, shelter belts

## Practice Questions

**Q1. Name the four main soils of India.**
*Answer:* Alluvial, black, red and laterite.

**Q2. Why is black soil good for cotton?**
*Answer:* It retains moisture for long, is rich in lime, magnesium and iron, and cotton needs a dry, warm season with moisture in the soil.

**Q3. Why is red soil red?**
*Answer:* It contains a lot of iron oxide produced by weathering of granite and gneiss.

**Q4. What is leaching and which soil does it produce?**
*Answer:* Heavy rain washes soluble minerals such as lime and silica downward, leaving iron and aluminium near the surface. It forms laterite soil.

**Q5. Differentiate between khadar and bhangar.**
*Answer:* Khadar is new, fine, fertile floodplain alluvium; bhangar is old, higher, clayey with kankar and less fertile.

**Q6. Mention two human causes of soil erosion.**
*Answer:* Deforestation and overgrazing (also faulty ploughing and jhum cultivation).

**Q7. Explain terrace farming.**
*Answer:* Steps are cut into hill slopes and levelled so that rainwater is held and run-off and erosion are reduced, giving flat fields for crops.

**Q8. Why is afforestation called the best soil conservation method?**
*Answer:* Roots bind soil, leaves break the fall of raindrops, and litter forms humus; wind and water speeds are reduced.

**Q9. Which soils are found in the Deccan Plateau?**
*Answer:* Black soil on the volcanic area, and red soil on the crystalline rocks to the east and south.

**Q10. What are shelter belts and where are they used?**
*Answer:* Rows of trees planted across the wind direction to reduce wind speed and prevent erosion, used in arid areas such as Rajasthan and Gujarat.

06

CHAPTER · GEOGRAPHY

# Natural Vegetation

Natural vegetation is what grows without human planting. Rain, temperature and altitude decide the type: dense rainforest in the wet west, thorny scrub in the desert and conifers on the high hills.



IN THIS CHAPTER

- Importance of forests
- Five vegetation types
- Mangroves
- Deforestation
- Conservation terms

## Why Forests Matter

- **Climate:** trees take in carbon dioxide, release oxygen and influence rainfall.
- **Soil and water:** roots hold the soil and let rain percolate to groundwater, reducing floods.
- **Resources:** timber, fuel wood, bamboo, medicinal herbs, lac, gum, honey, paper pulp.
- **Habitat:** homes for wildlife; they protect biodiversity.
- **Livelihood:** tribal communities and rural people depend on them.
- **Beauty and tourism:** national parks and sanctuaries.

## Types of Natural Vegetation

| TYPE | RAINFALL AND CLIMATE | WHERE | TREES | FEATURES |
|---|---|---|---|---|
| Tropical evergreen (rain forest) | Over 200 cm; hot, humid | Western Ghats, Assam, Meghalaya, Andaman and Nicobar | Mahogany, ebony, rosewood, rubber, bamboo | Tall (up to 60 m), dense, green all year with no fixed leaf-fall season; the thick canopy lets little light reach the ground |
| Tropical deciduous (monsoon) | 70 to 200 cm; clear dry season | Most of India: Madhya Pradesh, Uttar Pradesh, Chhattisgarh, Jharkhand, Odisha | Teak, sal, shisham, sandalwood, mulberry | Trees shed leaves in dry season to save water; most valuable timber forests |
| Tropical desert (thorn and scrub) | Below 70 cm; dry and hot | Rajasthan, Gujarat, parts of Deccan | Babul, kikar, khejri, acacia, cactus | Small thorny trees, long roots, waxy leaves |
| Littoral and swamp (mangrove) | Coastal, salty water | Sundarbans, Mahanadi, Godavari and Krishna deltas, Andaman | Sundari, mangrove, palm | Stilt roots above mud; breathing roots (pneumatophores) |
| Mountain | Falls with altitude; cold | Himalayas, Nilgiris | Oak, chestnut (to 2,000 m), deodar, pine, fir, spruce (1,500-3,000 m), alpine grass and moss higher up | Vegetation changes in layers with height, like going towards the poles |



SIDEKICK TIP

**Rainfall steps** from heavy to scanty: Evergreen (above 200), Deciduous (70-200), Desert (below 70). Check the rain in a question and you will know the forest.

FUN FACT

The Sundarbans, the largest mangrove forest in the world, is named after the *sundari* tree. It is also home of the Royal Bengal tiger, which is a good swimmer.

## Moist and Dry Deciduous

Moist deciduous forests (100-200 cm) have **teak** and **sal** in Odisha, Jharkhand and the Western Ghats foothills. Dry deciduous forests (70-100 cm) have **teak, shisham and tendu** on the plateau. Teak is used for furniture and building, sal for railway sleepers and bridges.

## Deforestation: Causes and Effects

- **Causes:** farming, mining, dams, roads and cities, fuel wood and timber, grazing and forest fires.
- **Effects:** soil erosion, floods and droughts, loss of wildlife, climate change, loss of livelihood.

## Forest Conservation Terms You Must Know

| TERM | MEANING |
|---|---|
| Afforestation | Planting trees on land that has not had forest cover for a long time or never had it |
| Reforestation | Replanting trees where forest has been cut or destroyed |
| Social forestry | Growing trees on community land, roadsides and canal banks for fuel, fodder and timber, with people's participation |
| Agroforestry | Growing trees together with crops or livestock on the same land |
| Van Mahotsav | Annual tree-planting festival in the first week of July, begun in 1950 by K. M. Munshi |

SIDEKICK TIP

**A**fforest = **A**dd new forest. **R**eforest = **R**eplace what was lost.

## Other Steps to Protect Forests

- Strict rules on felling and a National Forest Policy aiming at 33% of the land under forest.
- Protected areas: national parks, wildlife sanctuaries and biosphere reserves.
- Controlled grazing, fire watch, use of alternative fuels (LPG, biogas) to cut fuel wood demand.
- Public awareness, such as the Chipko movement in Uttarakhand, where villagers hugged trees to save them.

BOARD ALERT

Do not write 'desert vegetation needs a lot of water'. It stores water in stems and has deep roots. Also, the sal and teak forests are deciduous, not evergreen.

HOT QUESTION

The Western Ghats and Rajasthan lie at similar latitudes but their natural vegetation is totally different. Explain the difference using rainfall and temperature.

## Mind Map Recap

**Vegetation**

**Importance**
- Oxygen and rain
- Soil and water
- Timber and fuel
- Wildlife home

**Evergreen**
- Over 200 cm
- Mahogany, ebony, rosewood
- Western Ghats, North-East

**Deciduous**
- 70-200 cm
- Teak and sal
- Monsoon forests

**Desert**
- Under 70 cm
- Babul, kikar, cactus

**Littoral**
- Mangroves, sundari
- Sundarbans and deltas

**Mountain and conservation**
- Oak, deodar, pine, fir
- Afforestation, social forestry, agroforestry, Van Mahotsav

## Practice Questions

**Q1. Which type of forest covers the largest area in India?**
*Answer:* Tropical deciduous (monsoon) forest, found where rainfall is 70 to 200 cm.

**Q2. Why do deciduous trees shed leaves in the dry season?**
*Answer:* To reduce water loss by transpiration when moisture is scarce.

**Q3. Where are mangrove forests found and why are they special?**
*Answer:* In the Sundarbans (West Bengal), river deltas and the Andaman coast. They grow in salty tidal water with stilt and breathing roots, and protect coasts from storms and erosion.

**Q4. Mention four uses of forests.**
*Answer:* Timber, fuel wood, medicinal herbs, lac, gum, and protection of soil, climate and wildlife.

**Q5. What is social forestry?**
*Answer:* Raising trees on community and wasteland and along roads and canals with the people's involvement, to meet fuel, fodder and small timber needs.

**Q6. Differentiate between afforestation and reforestation.**
*Answer:* Afforestation plants trees on land that was not forest; reforestation replants where forest was destroyed.

**Q7. Why is vegetation in the Himalayas arranged in belts?**
*Answer:* Temperature falls as altitude rises, so each height has its own climate and plant type, from tropical forests at the foot to alpine meadows near the snow line.

**Q8. Name three trees of tropical evergreen forests and one product from each.**
*Answer:* Mahogany (furniture timber), rosewood (carved furniture), rubber (latex).

**Q9. What is the Chipko movement?**
*Answer:* A movement in which villagers of the Himalayas embraced trees to stop them being felled.

**Q10. State two harms caused by deforestation.**
*Answer:* Soil erosion and floods, loss of wildlife habitat (and increase in carbon dioxide).

07

CHAPTER · GEOGRAPHY

# Water Resources

Only a sliver of Earth's water is fresh and usable, and India has to share it between over a billion people. Learn where our water comes from, why it is running short and how it is stored and used.



IN THIS CHAPTER

- Surface and ground water
- Need to conserve
- Rainwater harvesting
- Irrigation methods
- Bhakra Nangal and Hirakud

## Sources of Water

| SOURCE | WHAT IT IS | EXAMPLES |
|---|---|---|
| Surface water | Water on the surface, filled by rain and melting snow | Rivers, lakes, ponds, tanks, reservoirs, canals |
| Ground water | Water that has seeped below the surface and fills the pores of rocks (aquifers) | Wells, tube wells, springs, hand pumps |

- India gets nearly all its water from **precipitation**, mostly in the four monsoon months.
- The **Ganga, Brahmaputra and Indus** carry the biggest share of surface water.
- **Groundwater** is the main source for drinking water in rural India and for irrigation in places such as Punjab, Haryana, Uttar Pradesh and Gujarat.

## Why We Must Conserve Water

- Population, farming and industry are all increasing demand.
- Rain is uneven in time and place, so some regions face drought each year.
- Over-pumping of groundwater is dropping the water table in Punjab, Haryana, Rajasthan and Delhi.
- Pollution of rivers by sewage and factory waste reduces usable water.
- Wasteful use: leaking pipes, flood irrigation, unmetered supply.

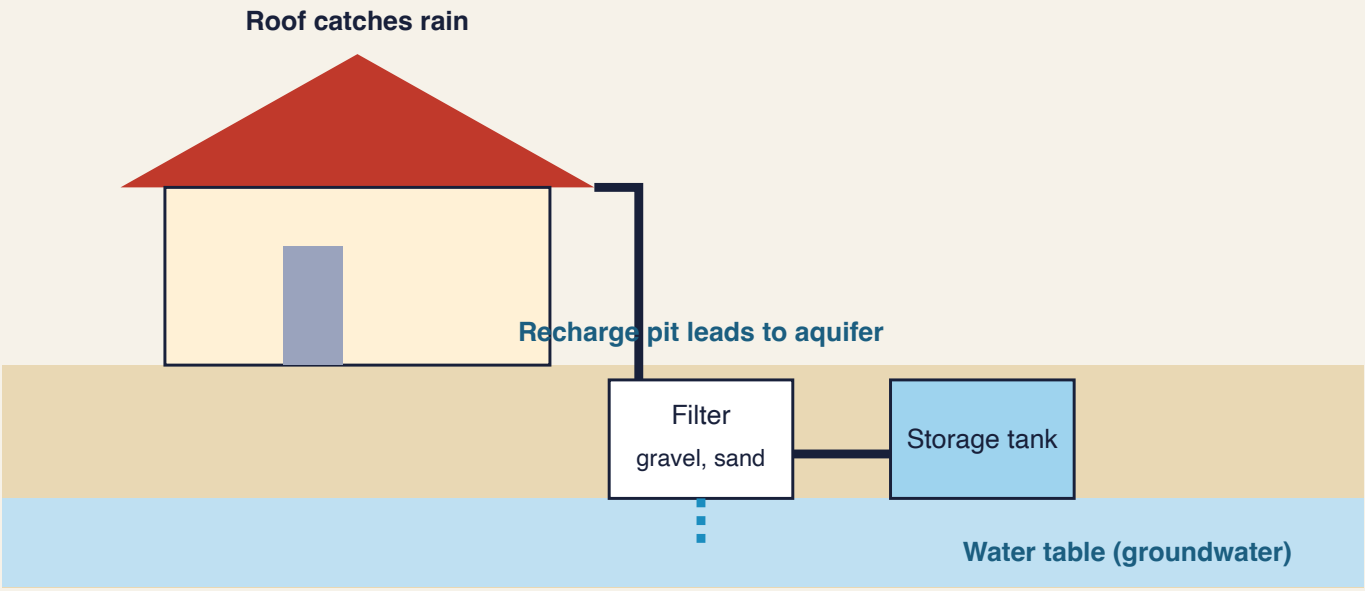


REAL LIFE

Mumbai gets over 200 cm of rain in four months, yet some areas of the city run short of water in summer, because storage and distribution matter as much as rain.

## Rainwater Harvesting

**Rainwater harvesting** means collecting and storing rain where it falls, instead of letting it flow away. It can be used at once or allowed to **recharge the groundwater**.




Roof-top rainwater harvesting: rain from the roof passes through a filter into a tank or a recharge pit.

| METHOD | HOW IT WORKS | WHERE |
|---|---|---|
| Roof-top harvesting | Rain from roofs is led through pipes into tanks or recharge pits | Houses, schools, apartments; compulsory in many cities |
| Check dams and percolation tanks | Small barriers across streams let water soak in | Rural India, Rajasthan, Gujarat, Maharashtra |
| Tankas and johads | Underground tanks and earthen ponds that store rain | Rajasthan |
| Bamboo drip, and kuls | Traditional channels bringing stream water to terraced fields | Meghalaya, Himalayas |

**Importance:** raises the water table, cuts flooding and run-off, lessens dependence on piped supply, helps agriculture in dry regions, costs little and is easy to maintain.

## Irrigation

**Irrigation** is the artificial supply of water to farmland. It is vital because the monsoon is uncertain and many areas get little rain, and because modern high-yielding seeds need regular water.

- It allows **more than one crop a year** and so increases production.
- It protects crops from drought and failure of the monsoon.
- It lets farmers grow water-hungry crops such as rice and sugarcane in drier areas.
- It supports high-yielding varieties and the Green Revolution.

| METHOD | HOW IT WORKS | WHERE IT IS COMMON | ADVANTAGES AND LIMITS |
|---|---|---|---|
| Wells and tube wells | Water is drawn from the ground by hand, an electric or diesel pump | Uttar Pradesh, Punjab, Gujarat, Rajasthan | Cheap, farmer's own, available all year; lowers the water table |
| Canals | Water from a river or reservoir is carried through channels | Northern plains, deltas of the Godavari and Krishna | Supply to large areas, long life; may cause waterlogging and salinity |
| Tanks | Rain or stream water collected in small ponds with a bund | Peninsular India: Tamil Nadu, Karnataka, Andhra Pradesh, Telangana | Suits uneven rock; dries in a drought, silts up |
| Drip irrigation | Pipes with tiny holes drop water at the root of each plant | Maharashtra, Karnataka, Tamil Nadu, Gujarat | Saves 40 to 70% of water, raises yield; costly to install |
| Sprinkler irrigation | Water is sprayed by a rotating nozzle like rain | Rajasthan, Haryana, Maharashtra | Good on sandy, uneven land; saves water; costly and loses some to wind |



SIDEKICK TIP

Remember the four land types: canals where rivers flow all year, tanks where the rock is hard and ground uneven, wells where groundwater is shallow, drip and sprinkler where water is scarce.

## Multi-purpose River Valley Projects

A dam serves several purposes at once: irrigation, electricity, flood control, drinking water, fisheries and tourism.

| | BHAKRA NANGAL | HIRAKUD |
|---|---|---|
| River | Satluj | Mahanadi |
| Location | Bhakra in Himachal Pradesh, Nangal in Punjab | Near Sambalpur, Odisha |
| Reservoir | Gobind Sagar | Hirakud reservoir, one of the largest artificial lakes in India |
| Type | Concrete gravity dam, one of the highest in the world (about 226 m) | Earthen dam with concrete and masonry sections, one of the longest dams in the world (about 4.8 km main dam) |
| Benefits | Irrigates Punjab, Haryana and Rajasthan; hydroelectricity for northern states; flood control; drinking water | Irrigates Odisha; hydroelectricity; controls floods in the Mahanadi delta; fishing and tourism |

KEY POINT

Bhakra Nangal = Satluj (north). Hirakud = Mahanadi (east). Both are **multi-purpose** projects.

## Problems of Large Dams

- People displaced and farmland and forests submerged.
- Sedimentation in reservoirs and loss of silt for downstream farmland.
- Waterlogging and salinity in canal command areas.
- Disputes between states over sharing river water.

HOT QUESTION

A village has plenty of rain but a falling water table. Suggest an irrigation method and a harvesting method to improve water availability and justify each.

## Mind Map Recap

**Water Resources**

**Sources**
- Surface: rivers, lakes, tanks
- Ground: wells, aquifers

**Need to save**
- Rising demand
- Uneven rain
- Falling water table
- Pollution

**Harvesting**
- Roof-top
- Check dams
- Tankas, johads
- Recharges aquifers

**Irrigation**
- Wells, tube wells
- Canals
- Tanks
- Drip and sprinkler

**Projects**
- Bhakra Nangal: Satluj
- Hirakud: Mahanadi
- Irrigation, power, floods

**Issues**
- Displacement
- Silting
- Salinity

## Practice Questions

**Q1. Distinguish between surface water and groundwater.**
*Answer:* Surface water is found in rivers, lakes and ponds on the land; groundwater lies below the surface in aquifers and is drawn through wells and tube wells.

**Q2. What is rainwater harvesting?**
*Answer:* Collecting and storing rainwater, either for direct use or to recharge groundwater.

**Q3. State three benefits of rainwater harvesting.**
*Answer:* It raises the water table, reduces floods and run-off, and gives a cheap supply of water in dry spells.

**Q4. Why is irrigation needed in India even in areas that receive good rainfall?**
*Answer:* Rain is concentrated in four months and varies from year to year, so farmers need water for dry months and for a second or third crop.

**Q5. Why are tanks common in the Peninsular plateau?**
*Answer:* The land is undulating with hard rock, so rainwater is easily trapped in hollows by bunds while wells and canals are difficult to build.

**Q6. What is drip irrigation? Give two advantages.**
*Answer:* Water is delivered slowly drop by drop to the roots through pipes. It saves water and reduces weeds and evaporation losses.

**Q7. Name the river and state on which Hirakud is built.**
*Answer:* The Mahanadi, in Odisha.

**Q8. State two benefits of the Bhakra Nangal project.**
*Answer:* It irrigates large areas of Punjab, Haryana and Rajasthan, and generates hydroelectricity (also controls floods).

**Q9. Why has groundwater fallen in Punjab and Haryana?**
*Answer:* Heavy use of tube wells for water-intensive crops such as paddy and wheat has taken out water faster than rain recharges it.

**Q10. Mention two ways to conserve water at home.**
*Answer:* Fix leaks and use a bucket instead of a shower; reuse kitchen water for plants; harvest rain from the roof.

08

CHAPTER · GEOGRAPHY

# Mineral and Energy Resources

Minerals are the raw material of industry and energy is its engine. Know where India's iron, copper, bauxite and manganese are mined, and compare coal, oil, gas, hydel, solar, wind, nuclear and the rest.



IN THIS CHAPTER

- Iron ore and manganese
- Copper and bauxite
- Coal, oil and gas
- Hydel power
- Non-conventional energy

## Minerals

A **mineral** is a naturally occurring substance with a definite chemical make-up. It is found in rocks and is mined. Minerals are **metallic** (iron, copper) and **non-metallic** (coal, mica). They are **non-renewable**, so they must be used wisely.

## Metallic Minerals: Uses and Distribution

| MINERAL | USES | TWO LEADING STATES AND MINING AREAS |
|---|---|---|
| **Iron ore** (haematite, magnetite) | Steel, machines, vehicles, ships, rails, buildings, construction | **Odisha**: Keonjhar, Mayurbhanj, Sundargarh (Barbil, Gorumahisani, Badampahar). **Chhattisgarh**: Bailadila in Bastar, Dalli-Rajhara in Durg. Also Jharkhand (Singhbhum: Noamundi, Gua, Kiriburu) and Karnataka (Bellary-Hospet, Kudremukh) |
| **Manganese** | Making steel hard and rust-resistant (ferro-manganese), dry cells, paints, glass, chemicals | **Madhya Pradesh**: Balaghat, Chhindwara. **Maharashtra**: Nagpur and Bhandara. Also Odisha (Keonjhar, Sundargarh) and Karnataka |
| **Copper** | Electric wires and cables, utensils, brass and bronze, electronics | **Madhya Pradesh**: Malanjkhand (Balaghat). **Rajasthan**: Khetri in Jhunjhunu. Also Jharkhand (Ghatsila in Singhbhum) |
| **Bauxite** (ore of aluminium) | Aluminium for aircraft, vehicles, cables, foil, utensils, cement | **Odisha**: Koraput, Kalahandi (Panchpatmali). **Gujarat**: Jamnagar, Kutch. Also Jharkhand, Maharashtra |

SIDEKICK TIP

Quick pairing: **Iron-Odisha-Chhattisgarh**; **Manganese-MP-Maharashtra**; **Copper-Rajasthan (Khetri)-MP (Malanjkhand)**; **Bauxite-Odisha-Gujarat**.

FUN FACT

Aluminium was once more precious than gold. Napoleon III is said to have served guests with aluminium cutlery while the others used gold.

## Energy Resources

**Conventional** sources are the traditional, mostly exhaustible ones: coal, petroleum, natural gas and hydel power (a renewable). **Non-conventional** sources are newer, renewable and generally clean.

## Conventional Sources

| SOURCE | DISTRIBUTION | ADVANTAGES | DISADVANTAGES |
|---|---|---|---|
| **Coal** | Jharkhand (Jharia, Bokaro, Giridih), West Bengal (Raniganj), Chhattisgarh (Korba), Odisha (Talcher), Madhya Pradesh (Singrauli). Lignite at Neyveli, Tamil Nadu | Cheap and plentiful; powers thermal plants and industry; raw material for coke and chemicals | Pollutes the air with carbon dioxide and sulphur; mining is dangerous; non-renewable |
| **Petroleum** | Mumbai High (offshore), Digboi, Naharkatiya and Moran in Assam, Ankleshwar and Cambay in Gujarat, Krishna-Godavari basin | Fuel for transport; base of petrochemicals and plastics; easy to move by pipeline | Imported in large quantity; pollution and oil spills; price swings; exhaustible |
| **Natural gas** | Krishna-Godavari basin, Mumbai High, Tripura, Jaisalmer in Rajasthan; the HBJ pipeline carries it from Hazira to Jagdishpur | Cleanest fossil fuel; used for power, fertilisers and as CNG and PNG | Needs costly pipelines; risk of leaks; non-renewable |

Coal types, in order of carbon content: **anthracite** (best), **bituminous**, **lignite**, **peat**. Most Indian coal is bituminous and is found in the Gondwana rocks of the Damodar valley.

## Hydel (Hydroelectric) Power

Electricity from falling water turns turbines in a power house at the foot of a dam. It is renewable and does not pollute, but it needs a steady flow and a good dam site.

| PROJECT | RIVER, STATE | BENEFITS |
|---|---|---|
| Bhakra Nangal | Satluj; Himachal Pradesh and Punjab | Cheap power to Punjab, Haryana, Rajasthan, Delhi and Himachal; irrigation; flood control |
| Hirakud | Mahanadi; Odisha | Power for Odisha and industries such as the Rourkela steel plant and aluminium smelting; irrigation; flood control for the delta |

## Non-Conventional Sources

| SOURCE | IMPORTANT AREAS | ADVANTAGES | DISADVANTAGES |
|---|---|---|---|
| **Solar** | Rajasthan (Bhadla), Gujarat, Karnataka, Tamil Nadu, Madhya Pradesh | Free, clean, inexhaustible; good for remote villages | High installation cost; works only in daytime and sunny weather |
| **Wind** | Tamil Nadu (Muppandal near Kanyakumari), Gujarat, Maharashtra, Karnataka, Rajasthan | Clean, cheap after installation, no fuel | Needs steady winds; noise; huge land; kills birds |
| **Tidal** | Gulf of Kutch, Gulf of Khambhat, Sundarbans | Predictable; no pollution | Few sites; very expensive; harms marine life |
| **Geothermal** | Manikaran (Himachal Pradesh), Puga valley (Ladakh) | Steady supply, small land use | Limited to hot spots; may release gases |
| **Nuclear** | Tarapur (Maharashtra), Rawatbhata (Rajasthan), Kalpakkam (Tamil Nadu), Narora (UP), Kakrapar (Gujarat), Kaiga (Karnataka), Kudankulam (Tamil Nadu) | Large output from little fuel; no smoke | Radioactive waste, accident risk, high cost; uranium (Jaduguda, Jharkhand) and thorium (Kerala sands) are needed |
| **Biogas** | Villages across India (gobar gas plants) | Cheap fuel and manure from dung and farm waste; clean cooking | Needs cattle and daily feeding; works poorly in cold |



KEY POINT

Renewable: hydel, solar, wind, tidal, geothermal, biogas. Exhaustible: coal, petroleum, natural gas. Nuclear fuel is limited, but needs little.



REAL LIFE

Solar panels on rooftops and wind turbines along the Tamil Nadu coast are examples of green energy that feed the national grid today.



BOARD ALERT

Do not write 'hydel power is a non-conventional source'. It is a conventional source, though renewable.



HOT QUESTION

India has huge coal reserves but is trying to move to solar and wind. Give three reasons why it should reduce its dependence on coal, and one reason why it cannot give it up quickly.

# Mind Map Recap

## Minerals and Energy

**Iron ore**
- Odisha, Chhattisgarh
- Steel, machines

**Manganese**
- MP, Maharashtra
- Hardens steel, dry cells

**Copper**
- MP (Malanjkhand), Rajasthan (Khetri)
- Wires

**Bauxite**
- Odisha, Gujarat
- Aluminium

**Fossil fuels**
- Coal: Jharia, Raniganj, Korba
- Oil: Mumbai High, Digboi
- Gas: KG basin

**Alternatives**
- Hydel: Bhakra, Hirakud
- Solar, wind, tidal, geothermal, nuclear, biogas

## Practice Questions

**Q1. Name two leading iron-ore producing states and one mining area in each.**
*Answer:* Odisha (Keonjhar or Mayurbhanj) and Chhattisgarh (Bailadila).

**Q2. State two uses of manganese.**
*Answer:* Making steel strong and rust-resistant (ferro-manganese alloy), and making dry-cell batteries.

**Q3. Which states are known for copper, and name the mining areas?**
*Answer:* Madhya Pradesh (Malanjkhand) and Rajasthan (Khetri-Singhana).

**Q4. Why is bauxite important?**
*Answer:* It is the ore of aluminium, a light, strong, rust-free metal used in aircraft, vehicles and cables.

**Q5. Mention one advantage and one disadvantage of coal.**
*Answer:* It is cheap and abundant for thermal power; it releases carbon dioxide and sulphur dioxide that pollute the air.

**Q6. Where is Digboi, and why is it famous?**
*Answer:* Digboi is in Assam. It is India's oldest oil field, with a refinery.

**Q7. Why is natural gas called a clean fuel?**
*Answer:* It burns almost without smoke or ash and gives far less carbon dioxide than coal or oil.

**Q8. State two benefits of Hirakud.**
*Answer:* It irrigates large areas of Odisha, generates hydroelectricity, and controls floods in the Mahanadi delta.

**Q9. Why is Rajasthan suitable for solar power?**
*Answer:* It has clear skies, long sunshine hours and large barren land.

**Q10. What is biogas and why is it useful in villages?**
*Answer:* A gas (mostly methane) formed by rotting dung and farm waste in a closed tank; it gives clean cooking fuel and rich manure.

09

CHAPTER · GEOGRAPHY

# Agriculture

Farming feeds India and employs nearly half its workers. Learn how farming is organised, when crops are sown, what each crop needs and where it grows, plus the story of the Green Revolution.



IN THIS CHAPTER

- Importance and problems
- Green Revolution
- Types of farming
- Kharif, rabi, zaid
- Major crops

## Importance of Agriculture

- It gives **food** to a very large population.
- About **45% of Indian workers** depend on it, and it contributes about one-sixth of GDP.
- It supplies **raw material** to industries: cotton, jute, sugarcane, oilseeds.
- It is a major source of **exports**: tea, spices, rice, sugar, cotton.
- It supports transport, trade and rural markets.

## Problems of Indian Agriculture

- **Dependence on monsoon** and uneven irrigation.
- **Small and fragmented holdings**, which make machines uneconomic.
- **Low use of modern inputs**: seed, fertilisers, mechanisation.
- **Soil exhaustion and erosion**; overuse of chemicals.
- Lack of **credit, storage and marketing**; farmers fall into debt, and middlemen take much of the price.

## Green Revolution

In the mid-1960s India faced food shortage. The **Green Revolution** was a package of reforms, beginning in 1966-67, in which **high-yielding variety (HYV) seeds** of wheat and rice, chemical fertilisers, pesticides, assured irrigation and machines were used to raise production sharply. **M. S. Swaminathan** led its Indian effort, with the Mexican scientist **Norman Borlaug** contributing the dwarf wheat.

| IMPACT OF THE FIRST GREEN REVOLUTION | DRAWBACKS |
|---|---|
| Foodgrain output more than doubled over the next two decades; India became self-sufficient in cereals | Mainly benefited **wheat and rice** and the regions of **Punjab, Haryana and western Uttar Pradesh**, leaving others behind |
| Incomes of farmers rose; rural jobs and agro-industry grew | Large farmers gained more than small ones |
| Cut dependence on imports and food aid | Falling water table, soil damage and pollution from chemicals |



FUN FACT

Dwarf wheat varieties were bred to have short, thick stems so that they would not bend under their heavy ears and could take plenty of fertiliser.

## Types of Farming

| TYPE | FEATURES | EXAMPLE |
|---|---|---|
| **Subsistence** | Produced mainly for the farmer's family; small farms, simple tools | Common in most Indian villages |
| **Commercial** | Produced for sale in the market; modern inputs | Sugarcane, cotton, jute, tea in many regions |
| **Intensive** | Small land, high inputs of labour and chemicals, several crops a year | Rice fields of the Ganga plain, deltas |
| **Extensive** | Large land, machines, lower yield per hectare but high total | Parts of Punjab, Haryana, Madhya Pradesh |
| **Plantation** | Single crop on large estates, with managers, labour and processing units, mainly for export | Tea (Assam), coffee (Karnataka), rubber (Kerala) |
| **Mixed** | Crops and animals on the same farm | Dairy with crops in Punjab, Gujarat |

## Agricultural Seasons

| SEASON | SOWING | HARVEST | CROPS |
|---|---|---|---|
| **Kharif** | June to July (start of monsoon) | September to October | Rice, jowar, bajra, maize, cotton, jute, groundnut, soyabean, sugarcane |
| **Rabi** | October to December | March to April | Wheat, barley, gram, peas, mustard, linseed |
| **Zaid** | March to June (between rabi and kharif) | Before the monsoon | Watermelon, muskmelon, cucumber, vegetables, fodder |



SIDEKICK TIP

**K**harif = **K**ome with the monsoon (June); **R**abi = **R**ain-less winter; **Z**aid = **Z**ips in between, in summer.

# Major Crops

## Food crops

| CROP | CLIMATE | SOIL | METHODS | LEADING STATES |
|---|---|---|---|---|
| **Rice** | Hot and humid, above 24°C; rain above 100 cm (150-200 cm ideal) | Alluvial or clayey that hold water | Seedlings raised and transplanted into flooded fields; HYV seeds, a lot of labour | West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh and Telangana, Tamil Nadu, Bihar, Odisha |
| **Wheat** | Cool growing season (10-15°C), warm ripening (21-26°C); rain 50-75 cm | Well-drained loam and clayey loam | Rabi crop; ploughing, sowing by seed drill, irrigation, mechanical harvest | Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan |
| **Millets** (jowar, bajra, ragi) | Warm, dry; rain 30-100 cm; hardy | Light, sandy, red and black soils, poor soils | Mostly rain-fed, grown with little care; kharif | Jowar: Maharashtra, Karnataka. Bajra: Rajasthan, Gujarat, Uttar Pradesh, Haryana. Ragi: Karnataka, Tamil Nadu |
| **Pulses** (gram, tur, moong, urad) | Moderate warm; 50-75 cm | Loamy, sandy loam, even poor soils | Rotation with cereals, because roots fix nitrogen | Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh |

## Cash and oilseed crops

| CROP | CLIMATE | SOIL | METHODS | LEADING STATES |
|---|---|---|---|---|
| **Sugarcane** | Hot and humid, 21-27°C; rain 75-150 cm; irrigation where dry | Deep, rich loam | Planted from stem cuttings; needs a lot of labour; mills near the fields to crush within 24 hours | Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu |
| **Groundnut** | Warm, 20-30°C; rain 50-75 cm | Light sandy or well-drained red soil | Kharif; sown at the start of the rains | Gujarat, Rajasthan, Tamil Nadu, Andhra Pradesh, Karnataka |
| **Mustard** | Cool, 10-25°C; rain 50-75 cm | Loam, sandy loam | Rabi; sown in October | Rajasthan, Haryana, Uttar Pradesh, Madhya Pradesh |
| **Soyabean** | Warm moist; 20-30°C; rain 60-100 cm | Well-drained loam or black soil | Kharif; often with rotation | Madhya Pradesh, Maharashtra, Rajasthan |
| **Cotton** | 21-30°C, at least 210 frost-free days; rain 50-100 cm | Black soil (regur) of the Deccan | Sown after the first rains; picked by hand | Gujarat, Maharashtra, Telangana, Andhra Pradesh |
| **Jute** (golden fibre) | Hot, humid; 24-35°C; rain over 150 cm | Well-drained alluvium, annually flooded | Stalks are cut, retted (soaked) in water for the fibre | West Bengal, Bihar, Assam |
| **Tea** | Warm, moist; 20-30°C; rain 150-300 cm, no frost | Deep, well-drained, acidic soil on slopes | Plantation farming; leaves plucked by hand; processed in factories on the estates | Assam, West Bengal (Darjeeling), Tamil Nadu (Nilgiris), Kerala |



KEY POINT

Cotton = black soil. Jute = golden fibre = West Bengal. Tea = slopes + no waterlogging. Rice needs the most water, wheat needs a cool winter.



REAL LIFE

Bengal's jute is used for gunny bags, carpet backing and ropes; tea from Assam is among the world's most traded.



BOARD ALERT

Do not write wheat as a kharif crop or rice as a rabi crop (rice is grown in rabi only in a few southern areas). Also keep millets apart from pulses: millets are cereals.



HOT QUESTION

Punjab grows wheat in winter, when rain is scarce, yet produces more than any other region per hectare. Explain how irrigation and the Green Revolution made this possible.

# Mind Map Recap

## Agriculture

### Importance
- Food, jobs, raw material
- Exports

### Problems
- Monsoon, small holdings
- Credit and storage

### Green Revolution
- HYV seeds, fertilisers, irrigation
- Punjab, Haryana, W. UP
- Swaminathan

### Types
- Subsistence, commercial
- Intensive, extensive
- Plantation, mixed

### Seasons
- Kharif: rice, cotton
- Rabi: wheat, mustard
- Zaid: melons

### Crops
- Rice, wheat, millets, pulses
- Sugarcane, oilseeds
- Cotton, jute, tea

## Practice Questions

**Q1. Why is agriculture called the backbone of the Indian economy?**
*Answer:* It employs nearly half the workforce, feeds the population and supplies raw material to industry and exports.

**Q2. Name any three problems faced by Indian farmers.**
*Answer:* Dependence on rain, small fragmented holdings, lack of credit and marketing (others: poor storage, soil exhaustion).

**Q3. What is the Green Revolution and which crops did it favour?**
*Answer:* A rise in farm production in the 1960s through HYV seeds, fertilisers, irrigation and machines, mainly in wheat and rice.

**Q4. Differentiate between intensive and extensive farming.**
*Answer:* Intensive farming uses small plots with very high inputs of labour and capital for a high yield per hectare; extensive farming uses large lands with machines and gives a lower yield per hectare.

**Q5. What is plantation farming? Give two examples.**
*Answer:* Growing one cash crop on large estates for export with processing factories at the site. Tea in Assam and rubber in Kerala.

**Q6. Name the three agricultural seasons and one crop from each.**
*Answer:* Kharif (rice), rabi (wheat), zaid (watermelon).

**Q7. Write three geographical conditions for the cultivation of rice.**
*Answer:* High temperature above 24°C, heavy rain (100-200 cm) or irrigation, and clayey or alluvial soil.

**Q8. Why is cotton grown in the Deccan plateau?**
*Answer:* It has black soil that holds moisture, a warm climate, moderate rain and a long frost-free season.

**Q9. Mention two leading states producing sugarcane and two conditions it needs.**
*Answer:* Uttar Pradesh and Maharashtra. It needs a hot, humid climate and deep fertile soil with plenty of water.

**Q10. Why is tea grown on slopes?**
*Answer:* Slopes drain water quickly; stagnant water would damage the roots.

10

CHAPTER · GEOGRAPHY

# Manufacturing Industries

Industries turn raw material into goods and farmers' produce into jobs and exports. Learn how industries are classified, which factors decide their location, and the big five in the syllabus: sugar, cotton, silk, iron and steel, petrochemicals and electronics.



IN THIS CHAPTER

- Classification
- Location factors
- Sugar, cotton, silk
- Iron and steel plants
- Petrochemicals
- Electronics

## Importance of Industries

- They turn raw materials into finished goods that are worth more.
- They give **employment** and raise income and living standards.
- They earn **foreign exchange** through exports and reduce imports.
- They help agriculture (fertiliser, tractors) and are the base of urban growth.
- They spread development through related services and small industries.

## Classification of Industries

| BASIS | TYPES | EXAMPLES |
|---|---|---|
| Size | **Large scale** (big capital, many workers)<br>**Small scale** (small capital, local market) | Iron and steel, cotton mills<br>Handloom, pottery, bakery |
| Raw material | **Agro-based** (from farm products)<br>**Mineral-based** (from minerals) | Sugar, cotton and silk textiles, jute<br>Iron and steel, cement, aluminium, petrochemicals |
| Ownership | **Public sector** (government)<br>**Private sector** (individuals or companies)<br>Joint sector (government and private) | SAIL, BHEL, ONGC<br>Tata Steel, Reliance, Infosys |

## What Decides Where an Industry Is Set Up?

| FACTOR | WHY IT MATTERS | EXAMPLE |
|---|---|---|
| Raw materials | Bulky, perishable or weight-losing raw material pulls industry close | Sugar mills near cane fields; steel plants near coal and iron ore |
| Power | Energy for machines | Coal fields, hydel power |
| Market | Customers nearby cut transport cost | Cotton mills near cities |
| Labour | Cheap, skilled workers | Surat textile, Bengaluru electronics |
| Transport | Rail, road and ports | Port-based steel at Visakhapatnam |
| Capital and policy | Banks and government incentives, special zones | Gujarat, Tamil Nadu |
| Water, land, climate | Needed for processing and cooling | Humid climate helps cotton spinning |

# Agro-based Industries

### 1. Sugar industry

- **Raw material:** sugarcane, which is bulky, loses weight and sugar content if not crushed within 24 hours. So mills stand near the cane fields.
- **Leading states:** Uttar Pradesh, Maharashtra (mostly cooperative mills), Karnataka, Tamil Nadu, Andhra Pradesh and Gujarat.
- **By-products:** molasses (for alcohol and ethanol), bagasse (fuel and paper), press mud (manure).
- **Problems:** seasonal working, low cane yield and old machinery, and price disputes.

### 2. Cotton textile industry

- India's oldest modern industry; the first mill was set up at Mumbai in 1854.
- **Centres:** Mumbai (the 'Cottonopolis'), Ahmedabad ('Manchester of India'), Surat, Coimbatore ('Manchester of the South'), Kanpur, Indore, Solapur.
- **Why Mumbai and Gujarat?** Cotton from the black soil nearby, humid air (threads do not snap), a port for machinery and exports, capital and labour.
- The **handloom** and **power-loom** sectors give most of the jobs.

### 3. Silk textile industry

- India is the second largest producer of silk in the world and the only producer of all four silks: **mulberry, tasar, eri and muga**.
- **Karnataka** (Mysuru, Bengaluru, Ramanagara) produces most mulberry silk. Others: Tamil Nadu (Kanchipuram), Uttar Pradesh (Varanasi), West Bengal (Murshidabad), Assam (muga), Jharkhand (tasar).
- Needs silkworm rearing (sericulture) on mulberry leaves, a skilled workforce and a mild climate.



SIDEKICK TIP

Sugar mills are **raw-material oriented** (cane decays), cotton mills are **market and port oriented**, steel plants are oriented to **coal and iron ore**.

# Mineral-based Industries

### 4. Iron and steel

An **integrated steel plant** does everything on one site: melting iron ore in a blast furnace, converting it to steel and rolling it into sheets, rails and bars. It needs **iron ore, coking coal, limestone, manganese, water and cheap transport**.

| PLANT | STATE | SECTOR AND NOTE |
|---|---|---|
| Tata Steel (TISCO), Jamshedpur | Jharkhand | Private; set up in 1907 by Jamsetji Tata; near Singhbhum iron ore and Jharia coal; Subarnarekha river water |
| Bhilai | Chhattisgarh | Public (SAIL); with Russian help; iron ore from Dalli-Rajhara |
| Rourkela | Odisha | Public (SAIL); first public sector plant with German help; ore from Sundargarh; power from Hirakud |
| Durgapur | West Bengal | Public (SAIL); British help; on the Damodar valley coal |
| Bokaro | Jharkhand | Public (SAIL); Russian help; coal from Bokaro |
| Visakhapatnam | Andhra Pradesh | Public (RINL); port-based plant on the coast using imported coking coal |

FUN FACT

Jamshedpur was named after its founder Jamsetji Tata. It was India's first planned industrial city.

### 5. Petrochemical industry

**Petrochemicals** are products made from crude oil and natural gas: plastics, synthetic fibres (polyester, nylon), synthetic rubber, fertilisers, detergents, dyes and paints.

- **Key centres:** Mumbai, Vadodara (Gujarat), Jamnagar (Gujarat), Haldia (West Bengal), Panipat (Haryana), Mathura (UP), Barauni (Bihar), Koyali.
- Large refineries are built near ports or oilfields, linked by pipelines.
- They give raw material to the plastic, textile, paint and fertiliser industries, which is why petrochemicals are called the 'mother industry'.

### 6. Electronics industry

It makes computers, mobiles, televisions, telecom equipment, medical and defence electronics and now semiconductors.

- **Key centres:** Bengaluru (called India's Silicon Valley), Hyderabad, Pune, Chennai, Noida and Gurugram, Mumbai, Mohali, Thiruvananthapuram.
- **Relevance today:** backbone of IT, communication, banking, health, defence and space; large employer for educated youth; earns exports.
- They need skilled engineers, good power, airports and clean land rather than raw material.

## Industry and Pollution

Factories can pollute air, water and land. Remedies include treating waste water, filters and scrubbers on chimneys, clean fuels, green belts, and strict law.

BOARD ALERT

Do not call Bhilai or Rourkela private-sector plants. Only Tata Steel (Jamshedpur) is private; the SAIL plants are public. Also do not write Visakhapatnam as raw-material oriented; it is port-based.

HOT QUESTION

Why were most integrated iron and steel plants set up in Chhota Nagpur plateau and the surrounding region, but one at Visakhapatnam far from the coalfields?

## Mind Map Recap

### Industries

**Types**
- Large and small
- Agro and mineral based
- Public, private, joint

**Location**
- Raw material, power, market
- Labour, transport, policy

**Agro**
- Sugar: UP, Maharashtra, Karnataka
- Cotton: Mumbai, Ahmedabad, Coimbatore
- Silk: Karnataka, Kanchipuram

**Steel**
- TISCO Jamshedpur, Bhilai, Rourkela
- Durgapur, Bokaro, Visakhapatnam

**Petrochemicals**
- Mumbai, Jamnagar, Haldia
- Panipat, Mathura, Barauni

**Electronics**
- Bengaluru, Hyderabad, Noida
- IT, communication, defence

## Practice Questions

**Q1. Define 'agro-based industry'. Give two examples.**
*Answer:* An industry that uses agricultural produce as its raw material. Sugar and cotton textile.

**Q2. Why are sugar mills located near cane fields?**
*Answer:* Sugarcane is bulky and its sugar content falls if it is not crushed within about 24 hours of cutting.

**Q3. Why is Mumbai called 'Cottonopolis'?**
*Answer:* It was the leading cotton textile centre of India with many mills, owing to nearby cotton, humid climate, a port and capital.

**Q4. Name the four varieties of silk produced in India.**
*Answer:* Mulberry, tasar, eri and muga.

**Q5. What is an integrated steel plant?**
*Answer:* A single plant that carries out all stages from smelting iron ore to the final rolled steel products.

**Q6. Name two public sector steel plants and the states they are in.**
*Answer:* Bhilai (Chhattisgarh) and Rourkela (Odisha); also Durgapur, Bokaro, Visakhapatnam.

**Q7. What are petrochemicals? Name two products.**
*Answer:* Chemicals obtained from crude oil or natural gas. Plastics and synthetic fibre (also fertiliser, detergents).

**Q8. Why is Bengaluru a major electronics hub?**
*Answer:* It has skilled engineers, institutes, favourable climate, government support and good air connectivity.

**Q9. State any three factors that decide the location of an industry.**
*Answer:* Raw material, power and labour (or market, transport).

**Q10. Distinguish between public sector and private sector industries.**
*Answer:* The public sector is owned and run by the government (SAIL); the private sector is owned by individuals or companies (Tata Steel).

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CHAPTER · GEOGRAPHY

# Transport

Transport carries goods, people and ideas across the country. It links farm to factory to market, and connects India to the world. Compare the four modes and learn the great road corridors.



IN THIS CHAPTER

- Importance
- Roadways
- Railways
- Airways
- Waterways

## Why Transport Matters

- It brings **raw materials to factories** and finished goods to markets.
- It supports **trade**, tourism and jobs.
- It binds the nation together and helps in relief during floods, droughts and wars.
- It encourages **regional development**, as areas with good transport develop faster.

The four modes are **roads, railways, airways and waterways**.

## 1. Roadways

India has one of the world's largest road networks. Roads reach nearly every village and carry most of the passenger and a large part of the goods traffic.

| FACTORS AFFECTING DEVELOPMENT | EFFECT |
|---|---|
| Terrain | Plains give cheap roads; hills, deserts and marshes make them costly |
| Climate | Heavy rain, floods and snow damage or block roads |
| Economic level | A rich region can afford more and better roads |
| Government policy | Funds, plans and rules (rural road schemes, highways programmes) |

| ADVANTAGES | DISADVANTAGES |
|---|---|
| Door-to-door service, flexible routes | Slow and costly for heavy goods over long distances |
| Cheap and quick for short distances | Accidents and traffic jams |
| Roads are cheaper to build than railways; reach hills and villages | Air and noise pollution; damage by floods and rains |
| Good for perishable goods and small loads | Needs constant repair |

### Types of Roads

- **Expressways:** wide, multi-lane, access-controlled roads for fast traffic with no crossings, such as the Mumbai-Pune Expressway and the Yamuna Expressway.
- **National Highways:** main roads connecting state capitals, major cities, ports and borders, built and maintained by the Central Government. Example: NH 44 from Srinagar to Kanyakumari.
- **Golden Quadrilateral:** a super-highway network linking the four metros: **Delhi, Mumbai, Chennai and Kolkata**. It cuts travel time and boosts trade and industry.
- **North-South and East-West Corridors:** the North-South runs from Srinagar to Kanyakumari; the East-West from Silchar (Assam) to Porbandar (Gujarat).
- **State highways, district roads and village roads:** link towns to the national network.

KEY POINT

Golden Quadrilateral = 4 metros. NS-EW = Srinagar to Kanyakumari and Silchar to Porbandar.

## 2. Railways

The first train in India ran in **1853** from Bori Bunder (Mumbai) to Thane, a distance of 34 km. Indian Railways is one of the largest networks in the world and the main carrier of heavy goods and long-distance passengers.

| FACTORS AFFECTING DEVELOPMENT | EFFECT |
|---|---|
| Relief | Plains, such as the Ganga plain, are ideal; mountains, deserts and swamps need bridges and tunnels |
| Population and economy | Dense population and industries create demand |
| Resources | Mines and industrial areas attract lines |
| Cost and policy | Heavy investment is required |

| ADVANTAGES | DISADVANTAGES |
|---|---|
| Cheap for bulk goods and long distances | Costly to build and maintain |
| Safe and dependable, fixed timetable | Fixed routes and stations; no door-to-door service |
| Good for passengers: comfortable, with berths | Needs road transport for the first and last mile |
| Less pollution per tonne than road or air | Difficult in hills and deserts |

India uses three **gauges**: broad (1.676 m), metre (1 m) and narrow (0.762 m or 0.610 m). Most traffic now runs on broad gauge. Mountain lines such as the Darjeeling Himalayan Railway are narrow gauge. The **Konkan Railway** links Mumbai and Mangaluru.

## 3. Airways

| ADVANTAGES | DISADVANTAGES |
|---|---|
| Fastest mode over long distance | Costly, only for the better-off or valuable goods |
| Reaches remote and rough areas such as Ladakh and the North-East | Limited carrying capacity |
| Quick relief in disasters; good for urgent and perishable goods | Affected by fog and storms |
| Saves time | Noise and carbon emissions; airports are expensive |

Major international airports: Delhi (Indira Gandhi), Mumbai (Chhatrapati Shivaji Maharaj), Kolkata, Chennai, Bengaluru, Hyderabad. Domestic airlines include Air India and IndiGo.

## 4. Waterways

| INLAND WATERWAYS | COASTAL AND OCEAN (SEA) WATERWAYS |
|---|---|
| Rivers and canals: National Waterway 1 (Ganga from Prayagraj to Haldia), NW 2 (Brahmaputra from Sadiya to Dhubri), NW 3 (West Coast Canal in Kerala), Buckingham Canal, Kerala backwaters | Along 7,500 km of coast and across the oceans. Major ports: **Kandla, Mumbai, Jawaharlal Nehru Port (Nhava Sheva), Marmagao, New Mangalore, Kochi, Tuticorin, Chennai, Visakhapatnam, Paradip, Haldia (Kolkata)** |

| ADVANTAGES | DISADVANTAGES |
|---|---|
| Cheapest for heavy and bulky goods | Slow |
| No road to build; pollution is low | Rivers dry or silt up; ports need dredging |
| Handles overseas trade (about 90% of India's trade by volume goes by sea) | Limited reach inland; risk from storms and floods |



FUN FACT

The Kolkata port is a 'river port' about 130 km upstream of the sea on the Hugli. A newer port at Haldia was built to help big ships.



SIDEKICK TIP

Use **Cheap-Slow-Heavy** for waterways, **Fast-Costly-Light** for airways, **Bulk-Long-Fixed** for railways and **Door-to-door-Short-Flexible** for roads.



BOARD ALERT

Do not give NS-EW the wrong end points. The East-West corridor starts at Silchar, not Guwahati, and ends at Porbandar, not Kandla.



HOT QUESTION

A company must send 5,000 tonnes of coal from Jharia to Mumbai, 1 tonne of medicine to Leh and 200 kg of flowers from Bengaluru to Delhi. Pick a mode of transport for each and justify.

## Mind Map Recap

**Transport**

**Roads**
- Expressways, NH
- Golden Quadrilateral
- NS-EW corridors
- Door-to-door

**Railways**
- 1853 Mumbai-Thane
- Three gauges
- Bulk and long distance

**Air**
- Fast, costly
- Reaches remote regions

**Water**
- Inland: NW-1, NW-2, NW-3
- Coastal: 12 major ports
- Cheapest for bulk

**Factors**
- Terrain, climate, economy, policy

**Compare**
- Cheap bulk: rail, water
- Urgent: air
- Short, flexible: road

## Practice Questions

**Q1. What is the Golden Quadrilateral?**
*Answer:* A network of super highways joining the four metros Delhi, Mumbai, Chennai and Kolkata, designed for fast, safe travel and trade.

**Q2. Name the terminals of the NS and EW corridors.**
*Answer:* North-South: Srinagar to Kanyakumari; East-West: Silchar to Porbandar.

**Q3. Give two advantages and two disadvantages of roadways.**
*Answer:* Advantages: door-to-door service and suitability for short distances. Disadvantages: accidents and pollution.

**Q4. Why are railways more developed in the northern plains?**
*Answer:* The land is flat and fertile, the population and industry are dense, and building cost is low.

**Q5. When and between which stations did the first train run in India?**
*Answer:* In 1853 between Bori Bunder (Mumbai) and Thane.

**Q6. State two advantages of airways.**
*Answer:* Speed and ability to reach remote and difficult areas.

**Q7. Name two inland waterways of India.**
*Answer:* National Waterway 1 on the Ganga and National Waterway 2 on the Brahmaputra.

**Q8. Why is water transport the cheapest?**
*Answer:* The sea and rivers are natural routes with no cost of making a road or track, and ships can carry large loads for little fuel.

**Q9. List any four major ports on the west coast.**
*Answer:* Kandla, Mumbai, Jawaharlal Nehru Port, Marmagao (also New Mangalore and Kochi).

**Q10. What is an expressway? Name one.**
*Answer:* A wide, multi-lane, controlled-access highway for fast traffic; the Mumbai-Pune Expressway.

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CHAPTER · GEOGRAPHY

# Waste Management

Every home, farm and factory makes waste. Left alone, it spoils the land, poisons the water and air, and spreads disease. Learn the terms examiners ask about and the methods that make waste useful.



IN THIS CHAPTER

- Impact of waste
- Key terms
- Segregation, dumping, composting
- Reduce, reuse, recycle

## What Is Waste?

**Waste** is anything we throw away because it is no longer wanted: kitchen scraps, paper, plastic, glass, metal, old electronics, hospital leftovers, factory residue and sewage.

| KIND | EXAMPLES |
|---|---|
| Domestic and municipal | Food scraps, paper, plastic, glass, garden waste |
| Industrial | Fly ash, slag, chemical sludge, hot water |
| Agricultural | Straw, husk, manure, pesticide containers |
| Biomedical | Used syringes, bandages, medicines |
| E-waste | Old phones, computers, batteries |

Waste is also **biodegradable** (broken down by microbes: food, paper, leaves) or **non-biodegradable** (plastics, glass, metal, e-waste).

## Impact of Waste Accumulation

- **Spoilage of landscape:** overflowing dumps and litter make cities ugly, block drains and spoil tourist places.
- **Pollution:** garbage contaminates **land** (toxic leachate), **water** (rivers, lakes) and **air** (smoke from burning, methane from dumps, bad smell).
- **Health hazards:** flies, rats and mosquitoes breed in piles and spread cholera, typhoid, dysentery, dengue and malaria; chemicals and e-waste lead to cancer, skin and lung disease.
- **Danger to animals:** cows, birds and sea creatures swallow plastic.
- **Fire and floods:** methane at landfills catches fire; clogged drains cause urban floods.



REAL LIFE

Large landfills near big cities have caught fire in summer, sending smoke over homes. This is a direct result of mixed waste that is rich in methane.

## Why Waste Management Is Needed

- To protect public health and the environment.
- To save resources and energy by reusing materials.
- To keep towns clean and attractive.
- To cut greenhouse gases and pollution.
- To create jobs and useful products such as compost and energy.

## Key Terms You Must Know

| TERM | MEANING | LINK TO WASTE |
|---|---|---|
| Global warming | Rise in the average temperature of the Earth due to greenhouse gases | Waste dumps release methane and burning waste releases carbon dioxide |
| Acid rain | Rain made acidic by sulphur dioxide and nitrogen oxides in the air | Burning of waste and fuels; damages crops, buildings and water bodies |
| Eutrophication | Enrichment of water with nutrients that causes algae to bloom and oxygen to fall, killing fish | Sewage, detergents and fertilisers in waste water |
| Biomagnification | Increase in concentration of toxic, non-degradable substances at each higher level of a food chain | Pesticides such as DDT and heavy metals in waste |
| Thermal pollution | Rise in water temperature due to hot water released from factories and power plants | Reduces dissolved oxygen and harms aquatic life |



SIDEKICK TIP

Eutrophication: **E**xtra nutrients, **E**xtra algae, **E**mpty oxygen.

## Methods of Safe Disposal

### 1. Segregation

Sorting waste at the source into streams so that each can be treated. A common scheme uses **green bins for wet (biodegradable) waste**, **blue bins for dry recyclable waste** (paper, plastic, metal, glass) and **red or black bins for hazardous and sanitary waste**. Segregation is the first and most important step.

### 2. Dumping and landfills

- **Open dumping**: waste is thrown on open land. It is cheap but causes smell, pests and pollution of groundwater.
- **Sanitary landfill**: waste is spread in layers in a lined pit, compacted and covered with soil daily. Leachate is collected and methane may be tapped for energy. It needs plenty of land away from homes.

### 3. Composting

Wet waste is rotted by microbes into rich manure called **compost**. In **vermicomposting**, earthworms speed up the process. The product improves soil and reduces the use of chemical fertilisers.

| METHOD | SUITS | RESULT | LIMITATION |
|---|---|---|---|
| Segregation | All waste | Clean streams for treatment | Needs public habit |
| Sanitary landfill | Non-recyclable dry waste, ash | Safe burial of waste | Needs land, leachate care |
| Composting | Food, garden and farm waste | Manure | Slow, needs space |
| Incineration | Biomedical and hazardous waste | Ash and heat | Air pollution if not controlled |

## The Three R's: Reduce, Reuse, Recycle

| STEP | MEANING | HOW |
|---|---|---|
| **Reduce** | Create less waste in the first place | Carry cloth bags, avoid single-use plastic, buy only what you need, go paperless |
| **Reuse** | Use again for the same or a new purpose | Refill bottles, pass on clothes and books, use jars as containers, repair old things |
| **Recycle** | Process waste into new products | Paper into new paper, plastic into granules, glass and metal melted; old phones sent to e-waste recyclers |



KEY POINT

Order of importance: **Reduce > Reuse > Recycle**. The best waste is the waste never created.



FUN FACT

Recycling one tonne of paper can save about 17 trees' worth of fibre and a great deal of water and power.



BOARD ALERT

Do not write 'burning is the best way to get rid of waste'. It releases toxic gases and ash, and is allowed only for special hazardous waste in controlled incinerators.



HOT QUESTION

A colony of 500 homes dumps all waste, including kitchen leftovers, plastic and old batteries, into one open pit. List three problems that will arise and design a plan using the three methods of disposal in the syllabus.

## Mind Map Recap

**Waste Management**

**Impact**
- Landscape spoiled
- Land, water, air pollution
- Disease

**Need**
- Health, environment
- Resources
- Clean towns

**Terms**
- Global warming, acid rain
- Eutrophication
- Biomagnification, thermal pollution

**Disposal**
- Segregation
- Dumping, landfill
- Composting

**3 R's**
- Reduce
- Reuse
- Recycle

**Waste types**
- Biodegradable and not
- Biomedical, e-waste

## Practice Questions

**Q1. Name two health hazards caused by uncollected garbage.**
*Answer:* Diseases such as cholera, typhoid, dengue and malaria, spread by flies, rats and mosquitoes breeding in the waste.

**Q2. What is meant by biomagnification?**
*Answer:* The rise in the concentration of a harmful non-degradable substance, such as DDT, at each higher step of a food chain.

**Q3. How does eutrophication affect a lake?**
*Answer:* Nutrients from sewage and fertilisers cause algae to grow thickly; when they die, bacteria use up oxygen and fish die.

**Q4. Why is segregation important?**
*Answer:* It separates wet, dry and hazardous waste so that composting and recycling can be done and landfill load is reduced.

**Q5. State two differences between open dumping and a sanitary landfill.**
*Answer:* In open dumping waste is left uncovered and pollutes air and water. A sanitary landfill uses a lined pit, daily soil cover and leachate and gas collection.

**Q6. What is vermicomposting?**
*Answer:* Making compost from organic waste with earthworms.

**Q7. Define thermal pollution with an example.**
*Answer:* A rise in water temperature because of hot water released by power plants or factories, reducing dissolved oxygen.

**Q8. Explain 'reduce, reuse and recycle' with one example each.**
*Answer:* Reduce: use cloth bags; reuse: refill a bottle; recycle: turn old paper into new paper.

**Q9. How does burning of waste contribute to acid rain?**
*Answer:* It releases sulphur dioxide and nitrogen oxides, which dissolve in rain to form acids.

**Q10. Why are plastics a long-term problem?**
*Answer:* They are non-biodegradable, remain for hundreds of years and harm animals and soil.

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CHAPTER · GEOGRAPHY

# Internal Assessment: Project Work

Twenty marks come from a project report on a topic you choose. They are won by planning, honest data and neat presentation, not by length. Here is the scheme, the topic list and a ready-made outline.



IN THIS CHAPTER

- Marking scheme
- Topic ideas
- Step-by-step plan
- Sample outlines

## How the 20 Marks Are Awarded

The project is judged by two examiners independently: your **subject teacher (10 marks)** and an **external examiner (10 marks)**, who is another teacher from your school. Each marks five criteria on a grade scale from Grade I (4 marks) to Grade V (0 marks).

| CRITERION | WHAT THE EXAMINER LOOKS FOR |
|---|---|
| Preparation | States objectives and defines the aspects to be studied, studies text and source material and lists them; uses correct geographical terms |
| Procedure / Testing | A sound method: surveys, interviews, maps, photographs, charts; variety of techniques and resourcefulness |
| Observation | Data recorded properly, supported by evidence; a coherent, concise account |
| Inference / Results | Analysis of the data and a clear conclusion that matches your aims |
| Presentation | Neat, with contents page, maps, diagrams, pictures, acknowledgements and bibliography in the right place |



KEY POINT

Five criteria x 4 marks each = 20 marks. Grade I = 4, II = 3, III = 2, IV = 1, V = 0.

## Choose One Topic

| AREA | SUGGESTED TOPIC |
|---|---|
| Local geography | Land use pattern in different regions of India (comparative study); survey of a local market: types of shops and services |
| Environment | Wildlife conservation efforts in India |
| Current issues | Development of tourism in India |
| Transport | Development of road, rail, water and air routes |
| Industry | Industries in your state: raw materials, procurement, wastes; classify as polluting or environment-friendly; ways to cut pollution |
| Economy | Need for industrialisation in India, latest trends and impact on the economy |
| Field visit | Water treatment plant, sewage plant, garbage dump or vermicompost site: how it works |

You may also pick your own topic under these broad areas.

## A Five-Step Plan That Works

**1. 1. Aim and plan**
Write 3 to 4 clear objectives and the limits of the study. Decide how you will collect data.

**2. 2. Collect**
Use **primary data** (survey, interviews, photographs, field notes) and **secondary data** (books, government sites, newspapers). List every source.

**3. 3. Organise**
Convert raw figures into tables, bar graphs, pie charts and sketch maps.

**4. 4. Analyse**
Compare, find patterns and explain causes using geography terms: relief, climate, resources, transport, population.

**5. 5. Present**
Contents, introduction, body, conclusion, bibliography and acknowledgement, neatly handwritten or typed.

## Sample Outline 1: Survey of a Local Market

- **Aim:** to study the types of shops and services in the market of _____ and how they meet people's needs.
- **Method:** walk the market and map each shop; count shops by type (grocery, clothes, medical, repair, food, electronics); interview 10 shopkeepers and 20 customers about where goods come from and how far people travel.
- **Data:** table of shop types, pie chart of shares, a sketch map with symbols, photographs.
- **Analysis:** which services dominate and why (population, location near roads, income); where goods come from; problems such as parking, waste and encroachment.
- **Conclusion:** link the findings to the objectives; suggest improvements.

## Sample Outline 2: Industries and Pollution

- Pick 4 industries in your state (e.g. sugar, textile, steel, chemicals). Make a table: raw material, source, mode of transport, product, waste generated, method of disposal.
- Classify each as polluting or environment-friendly and give reasons.
- Suggest ways to reduce pollution: filters, treatment plants, recycling, green belts, cleaner fuel, public awareness.
- Add a flow chart: raw material, process, product, waste.

## Sample Outline 3: Tourism in India

- Types: heritage, hill, beach, wildlife, pilgrimage, medical, adventure.

- Maps of the main centres with one picture and fact each.
- Benefits: jobs, foreign exchange, development of roads and hotels. Problems: waste, damage to environment, crowding.
- Suggestions: eco-tourism, waste rules, community involvement.

SIDEKICK TIP

Examiners reward **original data**. A neat table of 30 real survey answers beats three pages copied from the internet.

BOARD ALERT

Plagiarism costs marks. Write in your own words, name every source, and never paste text or pictures without acknowledgement.

HOT QUESTION

You are asked to study the impact of a nearby factory. Name three types of data you would collect, one method for each, and one table or diagram to present each.

## Mind Map Recap

**Project Work**

**Marks**
- 20 marks
- 10 teacher + 10 external

**Criteria**
- Preparation
- Procedure
- Observation
- Inference
- Presentation

**Topics**
- Local market, land use
- Wildlife, tourism, transport
- Industry, treatment plants

**Method**
- Primary and secondary data
- Maps, charts, photos

**Report**
- Contents, aim, body
- Conclusion, bibliography

**Do not**
- Copy
- Skip sources
- Hand in late

## Practice Questions

**Q1. How many marks are given for the geography project, and who awards them?**
*Answer:* 20 marks: 10 by the subject teacher and 10 by an external examiner.

**Q2. Name the five criteria used to assess the project.**
*Answer:* Preparation, procedure/testing, observation, inference/results and presentation.

**Q3. Differentiate between primary and secondary data. Give an example of each.**
*Answer:* Primary data are collected first-hand, such as a market survey; secondary data come from books or reports, such as census tables.

**Q4. Which tools can present data in a geography project?**
*Answer:* Tables, bar graphs, pie charts, sketch maps, flow charts and photographs.

**Q5. Why should the aim be stated clearly?**
*Answer:* It guides what to collect and analyse, and the conclusion has to match it; examiners check this link.

**Q6. List four items that must be present in a neat project file.**
*Answer:* Title page, contents, acknowledgement and bibliography (also maps, diagrams).

**Q7. Suggest two ways of reducing pollution from an industrial unit.**
*Answer:* Treating waste water before release and fitting filters on chimneys (or recycling waste).

**Q8. What should you do if you use information from a website?**
*Answer:* Rewrite in your own words and list the source in the bibliography.

# Done reading?
# Now go break the board.



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